Sinais e Sistemas para Engenharia

MÓDULO 15 - ANÁLISE DE EQUAÇÕES DE DIFERENÇAS

VERSÃO OCTAVE

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Um Sistema Discreto é normalmente descrito por uma Equação de Diferenças/ Equação de Recorrência / Algoritmo

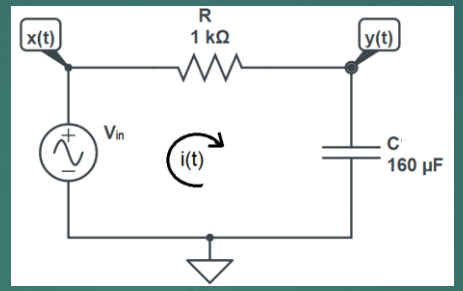
A) As Equações de Diferenças surgem normalmente da discretização de Equações Diferencias, com a escolha de uma taxa de amostragem adequada. Como no exemplos do Circuito elétrico de 1ª Ordem:

$$y'(t) + 6.25y(t) = 6.25x(t)$$

$$\frac{y[n] - y[n-1]}{T} + 6.25y(n) = 6.25x(t)$$

$$y[n] + 6.25\,h\,y[n] = y[n-1] + 6.25\,h\,x[n]$$

$$y(n) = \frac{1}{1 + 6.25T}(y[n-1] + 6.25\,T\,x[n])$$



$$y(t) \rightarrow y[n]$$

$$y'(t) \rightarrow y'[n] = \frac{y[n] - y[n-1]}{T}$$

$$x(t) \rightarrow x[n]$$

T=0.032

$$y[n] = \frac{5}{6}y[n-1] + \frac{1}{6}x[n]$$

B) As equações de diferenças podem ser especificadas diretamente no domínio discreto, como neste sistema que tira a média das últimas 3 amostras.

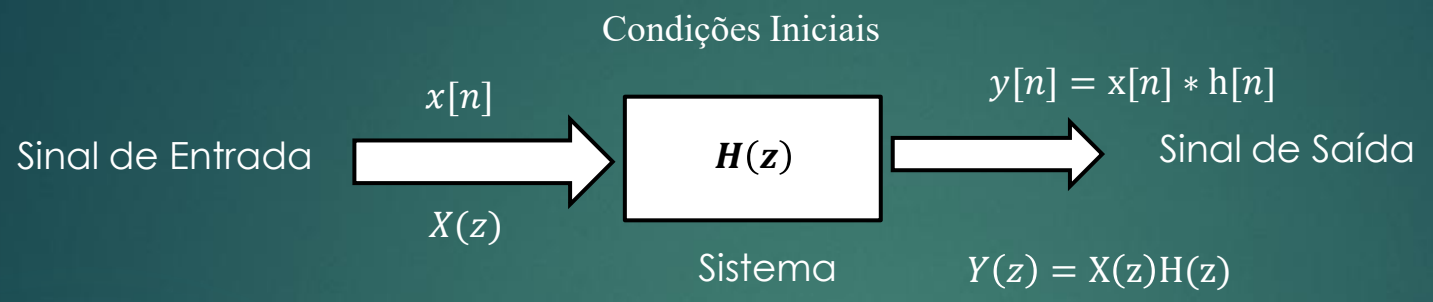
$$y\,(n) = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

C) O podem ser projetadas no domínio da Frequência (filtro passa baixas Chebychev) e transformadas para equações de diferenças, por exemplo, por Transformação bilinear, também com a escolha de uma frequência de amostragem adequada (que será estudada no próximo capítulo)

$$y[n] = \left(\frac{57}{40}\right)y[n-1] - \left(\frac{37}{33}\right)y[n-2] + \left(\frac{37}{33}\right)y[n-3] + \left(\frac{5}{49}\right)x[n] + \left(\frac{3}{34}\right)x[n-1] + \left(\frac{3}{34}\right)x[n-2] + \left(\frac{5}{49}\right)x[n-3]$$

Sistemas Discretos podem ser Resolvidos:

Seja um sistema discreto pode ser descrito por uma equação de diferenças. A sua resposta $y(n)$ do sistema à uma entrada $\mathrm{x(n)}$ qualquer pode a princípio ser obtida por 4 caminhos:



1) Analiticamente no tempo, resolvendo diretamente a Equações de Diferenças. Esta opção, embora matematicamente viável, será abandonada, pois é um processo laborioso, muito semelhante à resolução de equações diferenciais, que pode ser substituída pela opção 2 .

2) Analiticamente na Frequência pela : se X(z) for a transformada Z do sinal de entrada x(n), a resposta do sistema Y(z) pode ser obtida a partir da multiplicação de X(z) pela função de transferência H(z), isto é, $Y(z) = \mathrm{X(z)H(z)}$. Então $y[n]$ pode ser obtida também a partir da transformada z inversa de Y(z)

3) Numericamente pela **Equações de Diferenças** no tempo : A partir de H(q) podemos obter a equação de diferenças (algoritmo), de modo que a saída $y[n]$ pode ser obtida recursivamente através deste algoritmo.

4) Numericamente pela **Convolução no Tempo**: se $\mathrm{x}[n]$ for a entrada do sistema, e $h[n]$ a resposta ao impulso do sistema, a saída será a convolução de $\mathrm{x}[n]$ por $h[n]$, isto é : $y[n] = x[n] * h[n]$ (No entanto esta possibilidade depende de uma resposta ao impulso finita!!)

- A partir do próximo exemplo, vamos seguir um esquema padrão de resolução.

Exemplo A : Calcule Analiticamente no tempo a Respostas ao Degrau e ao Impulso do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

Resposta homogênea

$$y_h[n] = A\left(\frac{5}{6}\right)^n$$

Resposta Forçada

$x[n] = u(n) = 1 \quad n \geq 0$

$y_f[n] = Bu(n)$

$y_f[n] = 5/6 y_f(n-1) + 1/6\, x[n]$

$B = 5/6\, B + 1/6$

$B = 1$

Resposta Completa ao Degrau

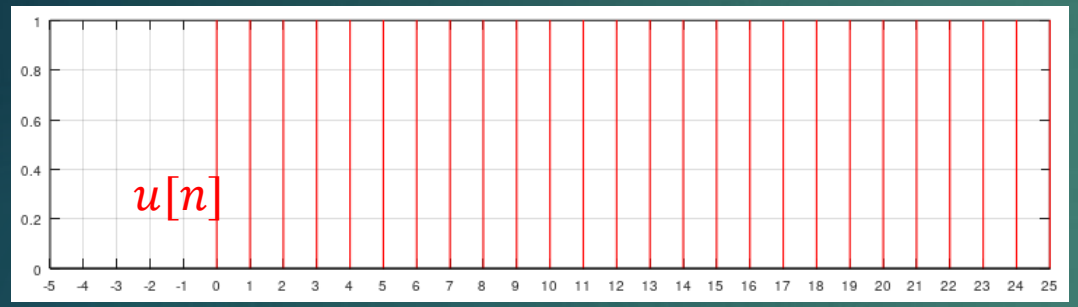
$y_o[n] = y_h[n] + y_f[n] = A(5/6)^n + 1$

Condições iniciais

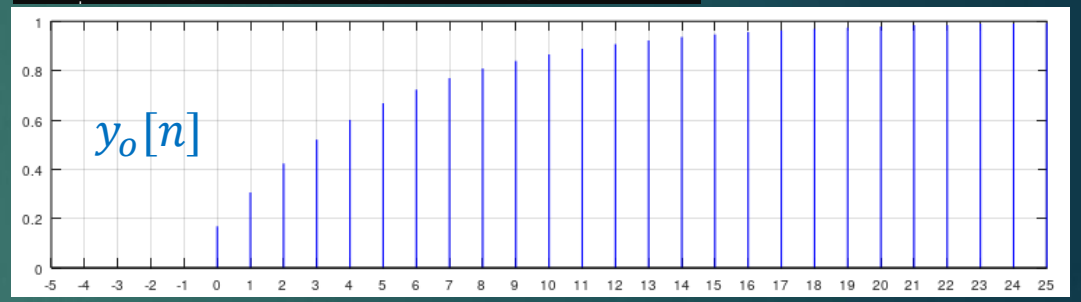
$y_o[-1] = 0 = A(5/6)^{-1} + 1 \rightarrow \quad A = -5/6$

$y_o[n] = 1 - 5/6(5/6)^n$

```
>> n=[-5:25];
>> stem(n,f_u(n),'r','marker','none')
```




```
>> f0 = @(n) (1-5/6*(5/6).^n).*f_u(n);
>> n=[-5:25];
>> stem(n,f0(n),'b','marker','none')
```

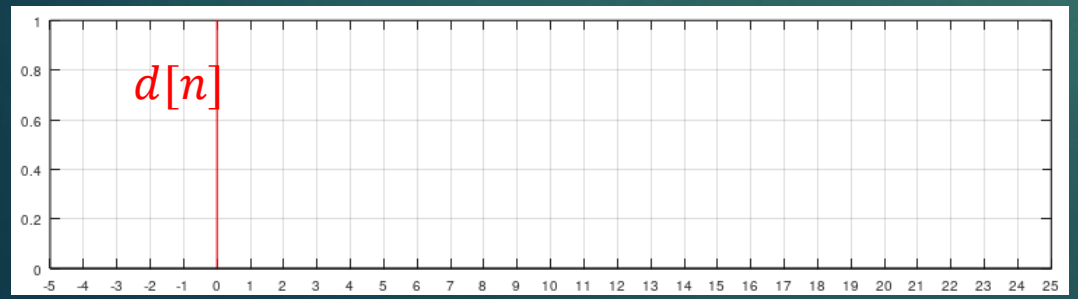


Resposta Completa ao impulso pode ser calculada diretamente da partir da primeira diferença da resposta ao degrau.

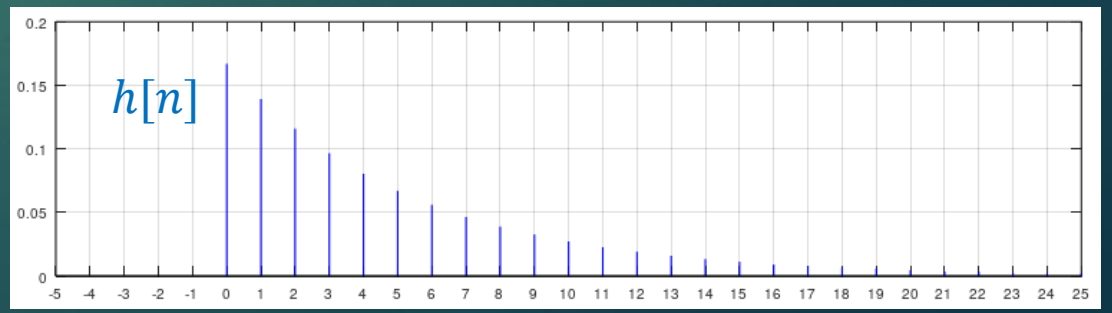
Como $\delta(n) = u[n] - u[n-1]$

$h[n] = y_o[n] - y_o[n-1] = 1 - 5/6(5/6)^n - (1 - 5/6(5/6)^{n-1}) = 1/6\,(5/6)^n$

```
>> n=[-5:25];
>> stem(n,f_d(n),'r','marker','none')
```




```
>> h = @(n) (1/6*(5/6).^n).*f_u(n);
>> n=[-5:25];
>> stem(n,h(n),'b','marker','none')
```



Calcule Analiticamente na Frequência as Respostas ao Degrau e ao Impulso do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

$$Y(z) = 5/6 z^{-1} Y(z) + 1/6 X(Z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{1}{6}z}{z - \frac{5}{6}}$$

```
>> n=[-5:25];
>> Hz=1/6*z/(z-5/6);
```

```
>> Xz=1;
>> fn=itransformadaZ_Hz(Xz*Hz,1,false)
fn =

@(n) (+(0.16667 * (0.83333) .^ n .* f_u (n)))

>> stem(n,fn(n),'b','marker','none')
```

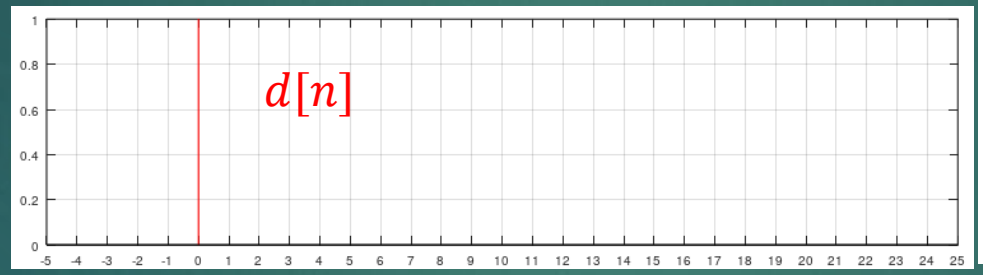
```
>> stem(n,f_d(n),'r','marker','none')
```

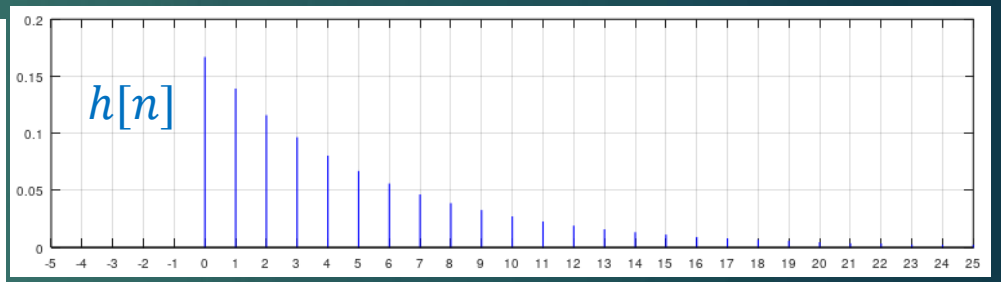
Resposta Completa ao Impulso

$X(z) = 1$

$$Y(z) = X(z)\, H(z) = \frac{\frac{1}{6}z}{z - \frac{5}{6}}$$

$$y[n] = \frac{1}{6}\left(\frac{5}{6}\right)^n$$






```
>> Xz=z/(z-1);
>> fn=itransformadaZ_Hz(Xz*Hz,1,true)
fn =

@(n) (+(1.00000 * f_u (n)) + (-0.83333 * (0.83333) .^ n .* f_u (n)))

>> stem(n,fn(n),'b','marker','none')
```

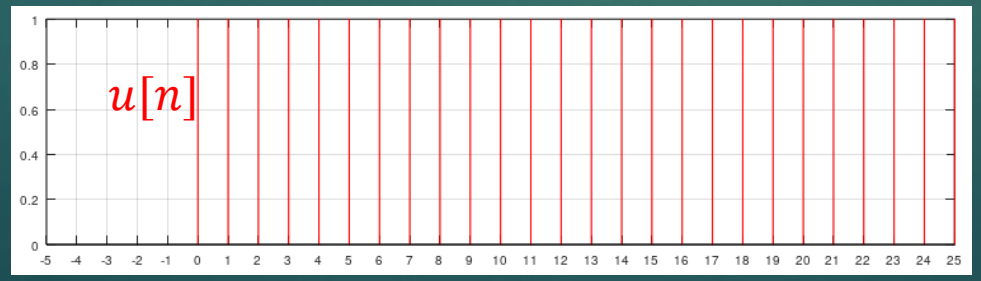
```
>> stem(n,f_u(n),'r','marker','none')
```

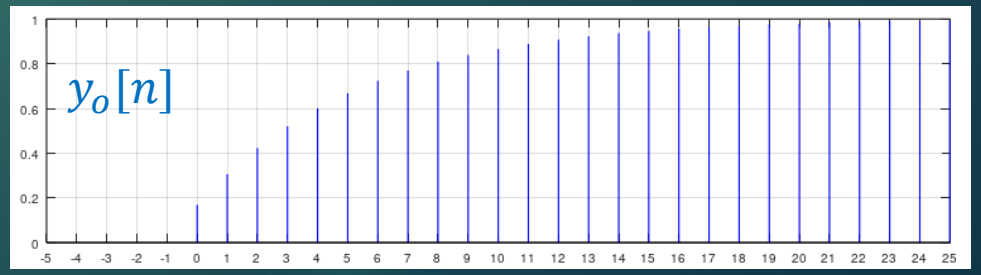
Resposta Completa ao Degrau

$$X(z) = \frac{z}{z-1}$$

$$Y(z) = X(z)\, H(z) = \frac{z}{z-1}\,\frac{\frac{1}{6}z}{z - \frac{5}{6}}$$

$$y_o[n] = 1 - \frac{5}{6}\left(\frac{5}{6}\right)^n$$





Calcule Analiticamente na Frequência a Resposta à entrada $x[n] = \left(\frac{1}{2}\right)^n$ com $y[-1] = 0.2$ do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{1}{6}z}{z - \frac{5}{6}}$$

```
>> n=[-5:25];
>> Hz=1/6*z/(z-5/6);
>> [num_c,den_c]=tfdata(Hz,'v');
```

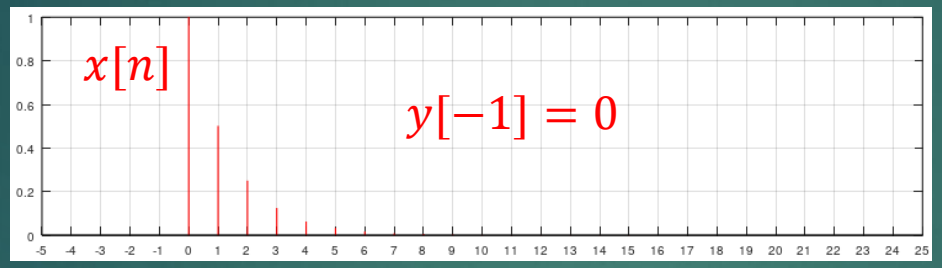
```
>> fx = @(n) (1/2).^n.*f_u(n);
>> stem(n,fx(n),'r','marker','none')
```

Resposta ao Estado Nulo

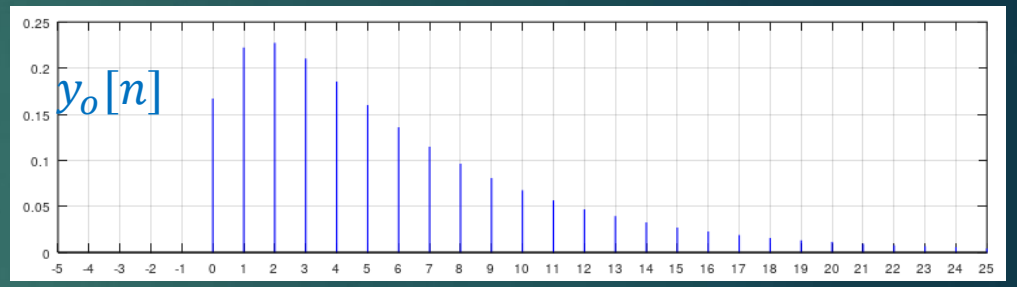
$$X(z) = \frac{z}{z - \frac{1}{2}}$$

$$Y_o(z) = X(z)\, H(z) = \frac{z}{z - \frac{1}{2}} \frac{\frac{1}{6}z}{z - \frac{5}{6}}$$

$$y_o[n] = \frac{5}{12}\left(\frac{5}{6}\right)^n - \frac{1}{4}\left(\frac{1}{2}\right)^n$$




```
>> Xz=z/(z-1/2);
>> fn=itransformadaZ_Hz(Xz*Hz,1,true)
fn =

@(n) (+(0.41667 * (0.83333) .^ n .* f_u (n)) + (-0.25000 * (0.50000) .^ n
 .* f_u (n)))

>> stem(n,fn(n),'b','marker','none')
```



Ordem 1
$y[n] = a_1\, y[n-1] + b_0\, x[n]$...
$X_n(z) = a_1\, y[-1]z$

Ordem 2
$y[n] = a_1\, y[n-1] + a_2\, y[n-2] + b_o\, x[n]$...
$X_n(z) = a_2 y(-1)z + \left(a_2 y(-2) + a_1 y(-1)\right)z^2$

Resposta Entrada Nula

$$X_n(z) = \frac{5}{6} y(-1)z = \left(\frac{5}{6}\right)(0.2)z$$

$$Y_n(z) = X_n(z)\frac{1}{Hz.den} = \frac{\frac{1}{6}z}{z - \frac{5}{6}}$$

$$y_n[n] = \frac{1}{6}\left(\frac{5}{6}\right)^n$$

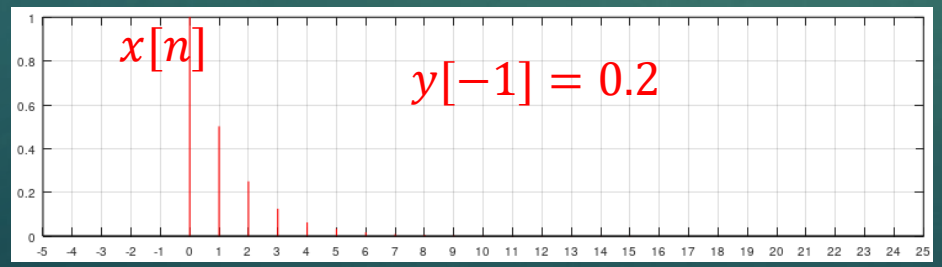
```
>> Xn=(5/6)*(0.2)*z;
>> Hn=tf(1,den_c,1);
>> fn=itransformadaZ_Hz(Xz*Hz+Xn*Hn,1,true)
fn =

@(n) (+(0.58333 * (0.83333) .^ n .* f_u (n)) + (-0.25000 * (0.50000) .^ n .* f_u (n)))

>> stem(n,fn(n),'b','marker','none')
```

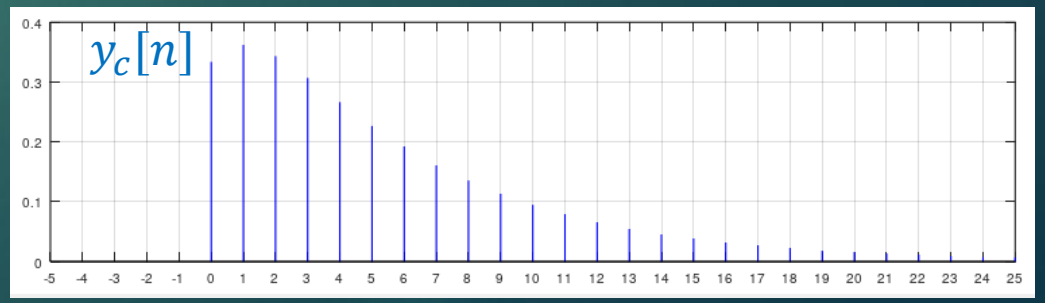
```
>> fx = @(n) (1/2).^n.*f_u(n);
>> stem(n,fx(n),'r','marker','none')
```

Resposta Completa

$$Y(z) = X(z)\, H(z) + X_n(z)\frac{1}{Hz.den}$$

$$y_c[n] = -\frac{1}{4}\left(\frac{1}{2}\right)^n + \frac{7}{12}\left(\frac{5}{6}\right)^n$$





Calcule Numericamente pela **Equações de Diferenças** a Resposta ao Impulso e ao Degrau do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

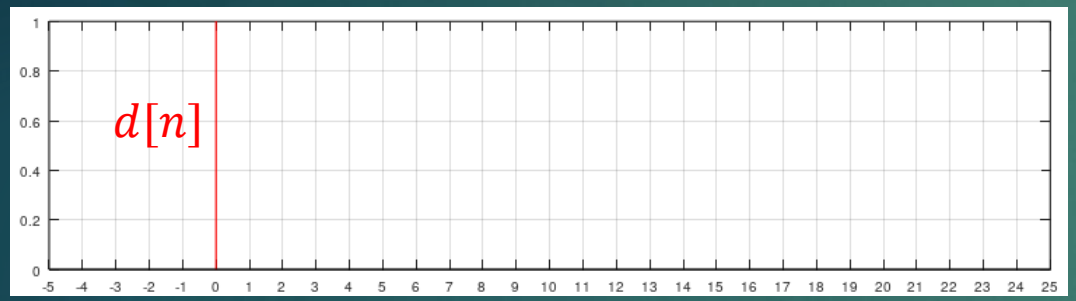
$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{1}{6}z}{z - \frac{5}{6}} = \frac{\frac{1}{6}}{1 - \frac{5}{6}z^{-1}}$$

Módulos 4
Sistemas Discretos – Equações de Diferenças

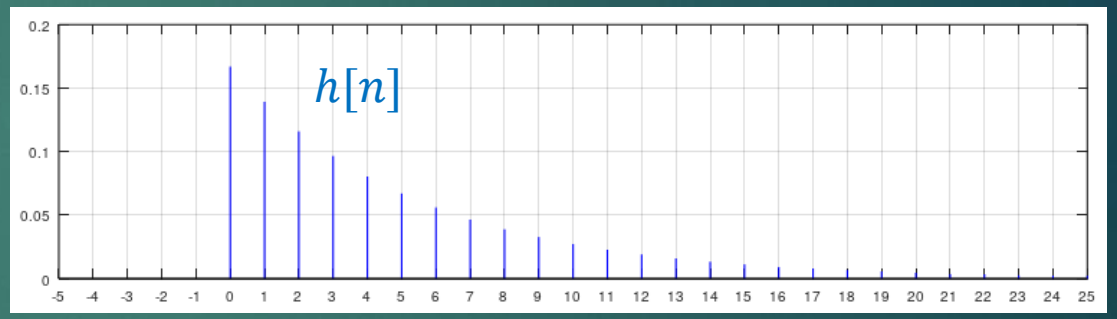
```
function y_out=Sistema_Eletrico_1aOrdem(x_in,y_ini)
    x=[0 x_in];
    y=[y_ini];
    for (i=2:length(x))
        y(i)=5/6*y(i-1)+1/6*x(i);
    end
    y_out=y(2:length(x));
endfunction
```

```
>> n=[-5:25];
>> x_in=f_d(n);
>> stem(n,x_in,'r','marker','none')
```

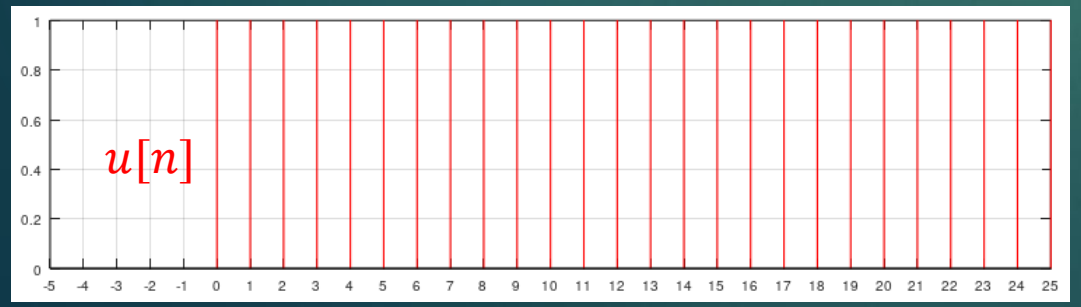



```
>> y_out=Sistema_Eletrico_1aOrdem(x_in,0);
>> stem(n,y_out,'b','marker','none')
```

esposta ao Impulso

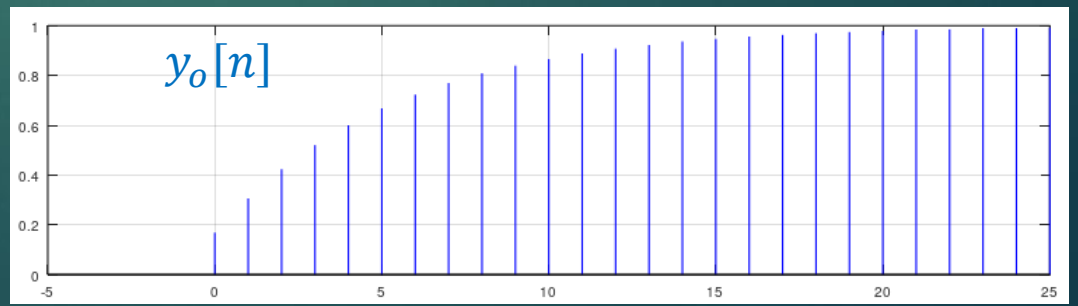



```
>> n=[-5:25];
>> x_in=f_u(n);
>> stem(n,x_in,'r','marker','none')
```




```
>> y_out=Sistema_Eletrico_1aOrdem(x_in,0);
>> stem(n,y_out,'b','marker','none')
```

Resposta ao Degrau



Algoritmo geral para resolver equações de diferenças

```
function y_out=applyIIR(Hz,x,xp,yp)
    [b,a]=tfdata(Hz,'v');
    npolos=length(a)-1;
    nzeros=length(b)-1;
    N=length(x);
    for n=1:N
      y2(n)=0;
      for np=1:npolos
          index=n-np;
          if index<=0
              y2(n)=y2(n)-a(np+1)*yp(npolos+index);
          else
              y2(n)=y2(n)-a(np+1)*y2(index);
          end
      end
      for nz=0:nzeros
          index=n-nz;
          if index<=0
               y2(n)=y2(n)+b(nz+1)*xp(nzeros+index);
          else
               y2(n)=y2(n)+b(nz+1)*x(n-nz);
          end
      end
     end
endfunction
```

$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

$$Y(z) = \frac{5}{6} z^{-1} Y(z) + \frac{1}{6} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1/6}{1 - \frac{5}{6} z^{-1}}$$

$$\text{com } \mathrm{q} = \mathrm{z}^{-1}$$

$$\mathrm{H(q)} = \frac{1/6}{1 - \frac{5}{6} q}$$

Notar que os coeficientes da equação de diferenças é obtido diretamente de $\mathrm{H(q)}$

```
function y_out=Sistema_Eletrico_1aOrdem(x_in,y_ini)
    x=[0 x_in];
    y=[y_ini];
    for (i=2:length(x))
          y(i)=5/6*y(i-1)+1/6*x(i);
    end
    y_out=y(2:length(x));
endfunction
```

Calcule Numericamente pela **Equações de Diferenças** a Resposta ao Impulso e ao Degrau do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

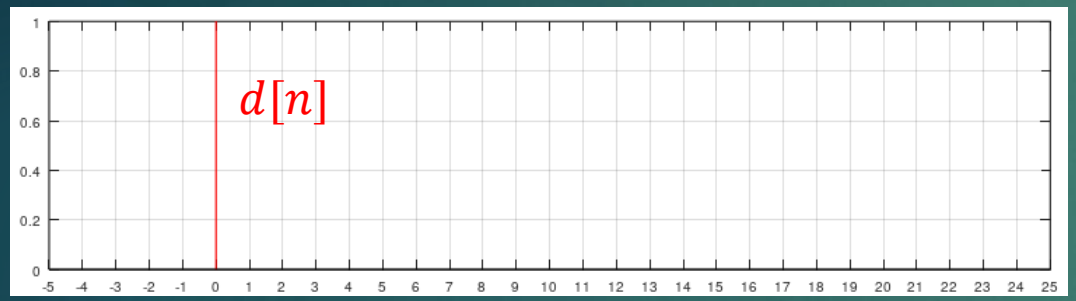
$$H(q) = \frac{1/6}{1 - \frac{5}{6} z^{-1}}$$

```
>> Hz=(1/6)/(1-5/6*z^-1);
>> [b,a]=tfdata(Hz,'v');
>> n=[0:25];
```

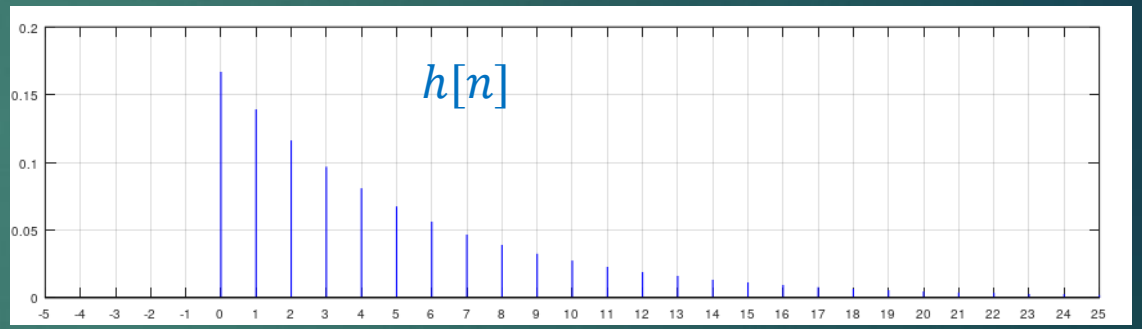
```
function y_out=applyIIR(Hz,x,xp,yp)
    [b,a]=tfdata(Hz,'v');
    npolos=length(a)-1;
    nzeros=length(b)-1;
    N=length(x);
    for n=1:N
      y2(n)=0;
      for np=1:npolos
      for nz=0:nzeros
    end
endfunction
```

Resposta ao Impulso

```
>> x_in=f_d(n);
>> stem(n,x_in,'r','marker','none')
```

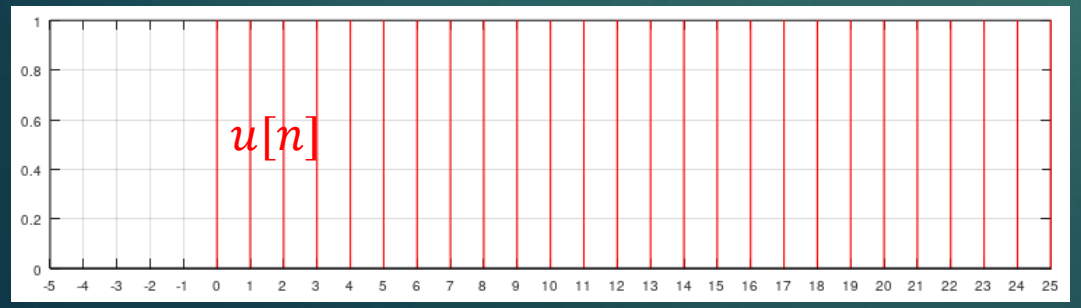



```
>> y_out=applyIIR(Hz,x_in,0,0);
>> stem(n,y_out,'b','marker','none')
```

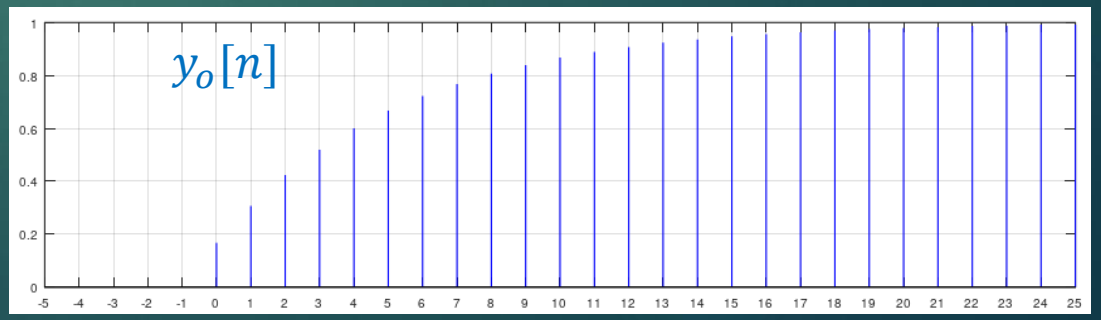


Resposta ao Degrau

```
>> x_in=f_u(n);
>> stem(n,x_in,'r','marker','none')
```




```
>> y_out=applyIIR(Hz,x_in,0,0);
>> stem(n,y_out,'b','marker','none')
```



Calcule Numericamente pela **Equações de Diferenças** a Resposta ao de Degrau, e à entrada $x[n] = \left(\frac{1}{2}\right)^n$ com $y(-1) = 0.2$ do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

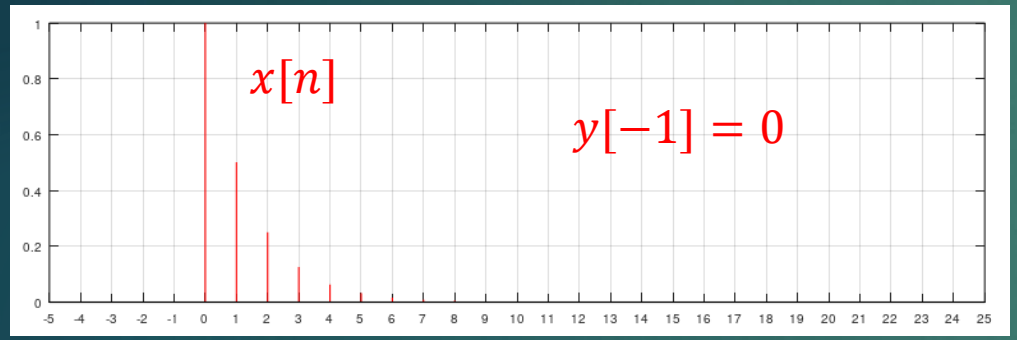
$$y[n] = 5/6\, y[n-1] + 1/6\, x[n]$$

$$H(q) = \frac{\frac{1}{6}}{1 - \frac{5}{6}q} \quad ; \quad com\ \ q = z^{-1}$$

```
>> Hz=(1/6)/(1-5/6*z^-1);
>> [b,a]=tfdata(Hz,'v');
>> n=[0:25];
```

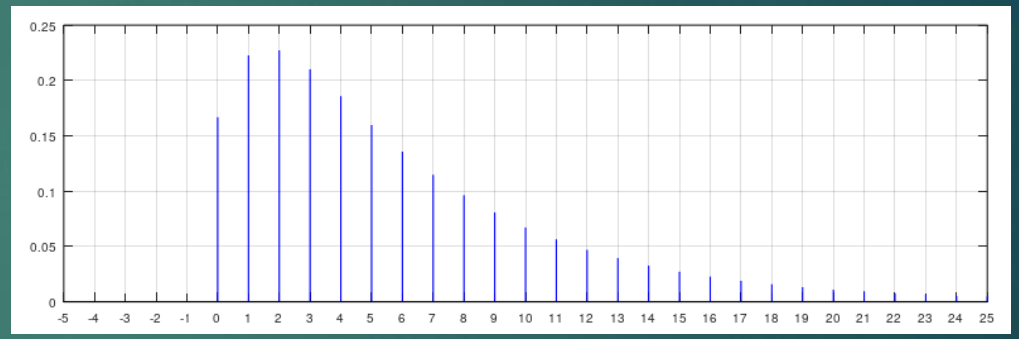
```
function y_out=applyIIR(Hz,x,xp,yp)
    [b,a]=tfdata(Hz,'v');
    npolos=length(a)-1;
    nzeros=length(b)-1;
    N=length(x);
    for n=1:N
      y2(n)=0;
      for np=1:npolos
      for nz=0:nzeros
    end
endfunction
```

```
>> fx = @(n) (1/2).^n.*f_u(n);
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```

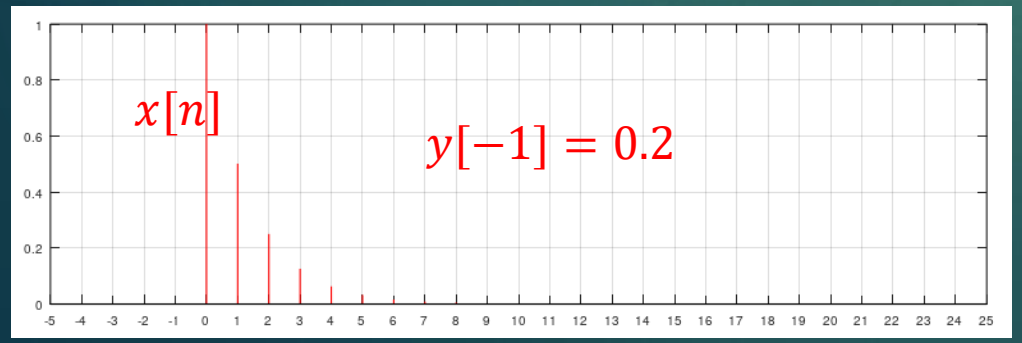



```
>> y_out=applyIIR(Hz,x_in,0,0);
>> stem(n,y_out,'b','marker','none')
```

Resposta ao estado nulo

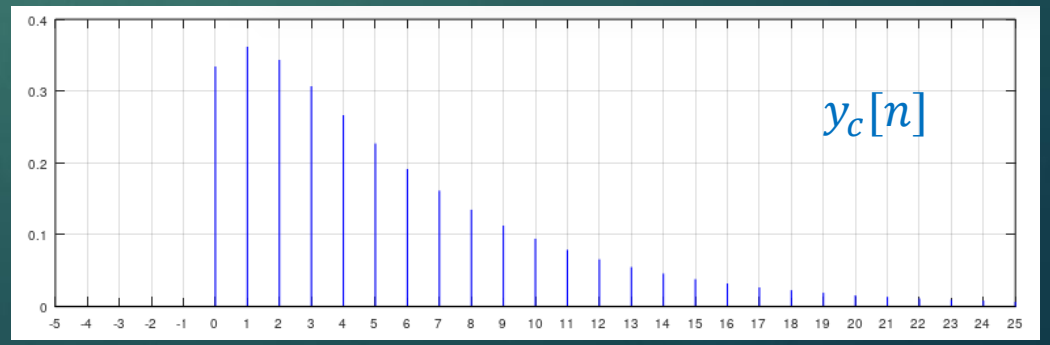


```
>> fx = @(n) (1/2).^n.*f_u(n);
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```




```
>> y_out=applyIIR(Hz,x_in,0,0.2);
>> stem(n,y_out,'b','marker','none')
```

Resposta Completa



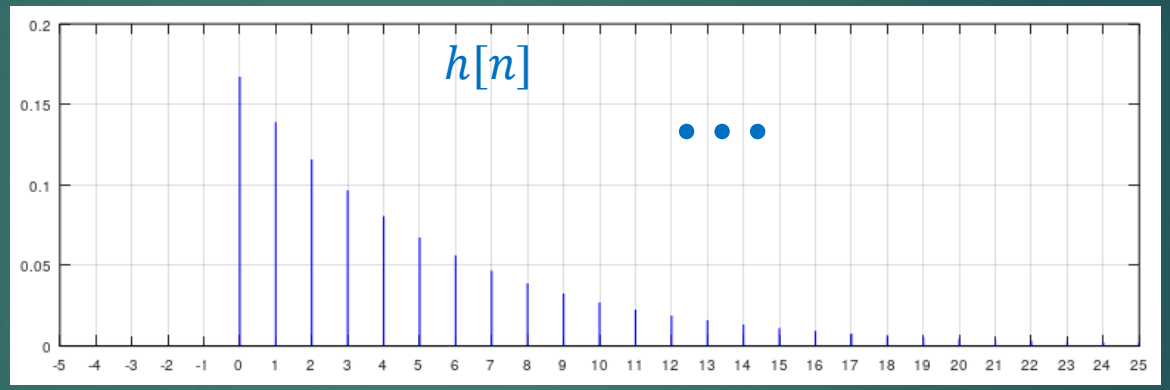
Resolução por **Convolução no Tempo** do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

$$y[n] = x[n] * h[n]$$

Precisamos primeiro calcular a resposta ao impulso do sistema $h(n)$. Podemos fazê-lo, Analiticamente no tempo, Analiticamente na Frequência, ou numericamente por Equações de diferenças.

$$h[n] = \frac{1}{6}\left(\frac{5}{6}\right)^n$$

```
>> fh = @(n) (1/6*(5/6).^n).*f_u(n);
>> n=[0:25];
>> h=fh(n);
>> stem(n,h,'b','marker','none')
```



- Como h(n) tem duração infinita, a princípio não podemos calcular a saída $y(n)$ por Convolução.
- Há técnicas utilizadas em filtros FIR que truncam $h(n)$ após um certo número de amostras, quando o valor de $h(n)$ se torna muito pequeno. Com este truncamento, a Convolução será uma aproximação, que será tanto melhor quanto maior for o número de amostras que preservarmos de h(n)
- Para isso, podemos simplesmente truncar $h(n)$, considerando que $h(n)\sim 0$ para n>25. Há outras técnicas de truncamento mais eficientes.
- Notar que com convolução, sempre obtemos a resposta ao estado nulo

Calcule Numericamente pela **Convolução no Tempo** a Resposta à entrada $x[n] = \left(\frac{1}{2}\right)^n$ com $y[-1] = 0$ do Sistema Discreto descrito pela Equação de Diferenças $y[n] = 5/6\, y[n-1] + 1/6\, x[n]$

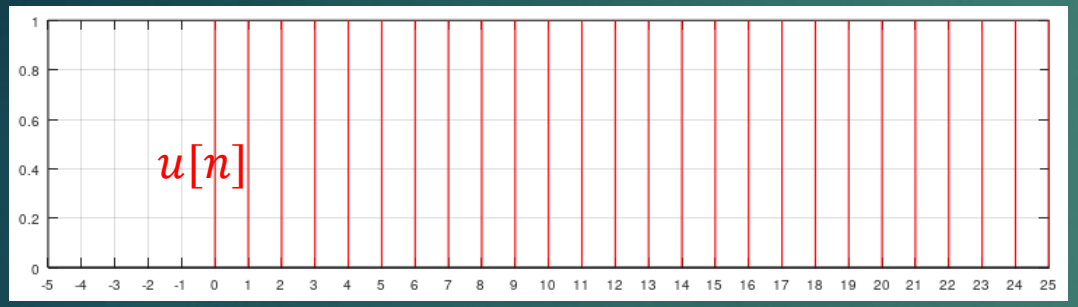
$$h[n] = \frac{1}{6}\left(\frac{5}{6}\right)^n$$

Truncado em N=26

```
>> fh = @(n) (1/6*(5/6).^n).*f_u(n);
>> n=[0:25];
>> N=length(n);
>> h=fh(n);
```

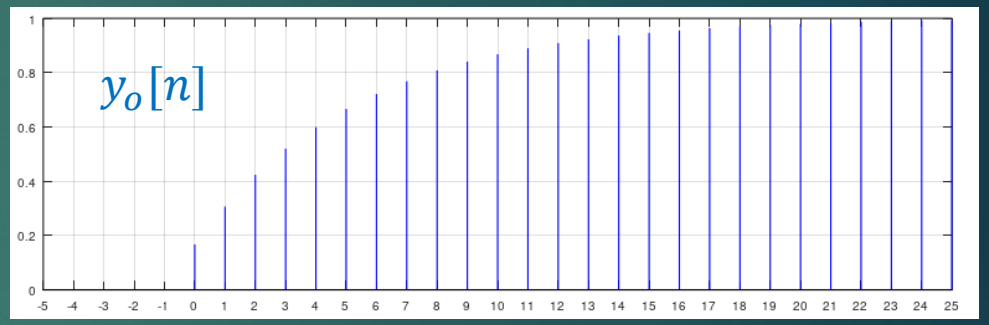
- Esta é somente uma aproximação, pois h(n) tem duração infinita, mas foi truncada com N=26.
- Notar que com convolução, sempre obtemos a resposta ao estado nulo

```
>> x_in=f_u(n);
>> stem(n,x_in,'r','marker','none')
```

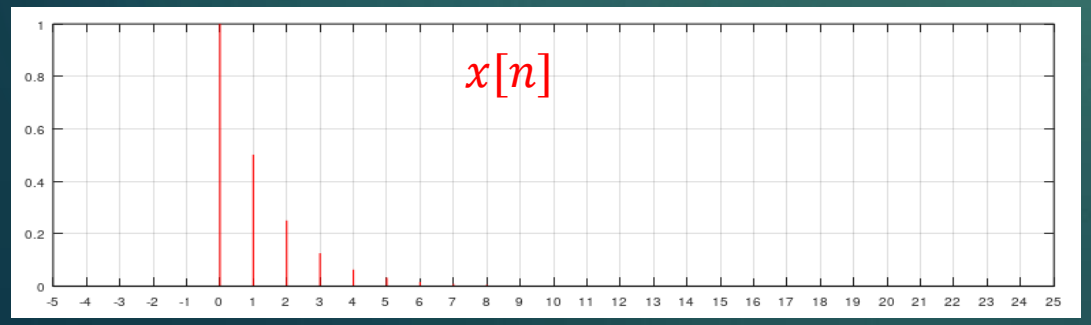


Resposta ao estado nulo

```
>> y_out=conv(x_in,h);
>> stem(n,y_out(1:N),'b','marker','none')
```

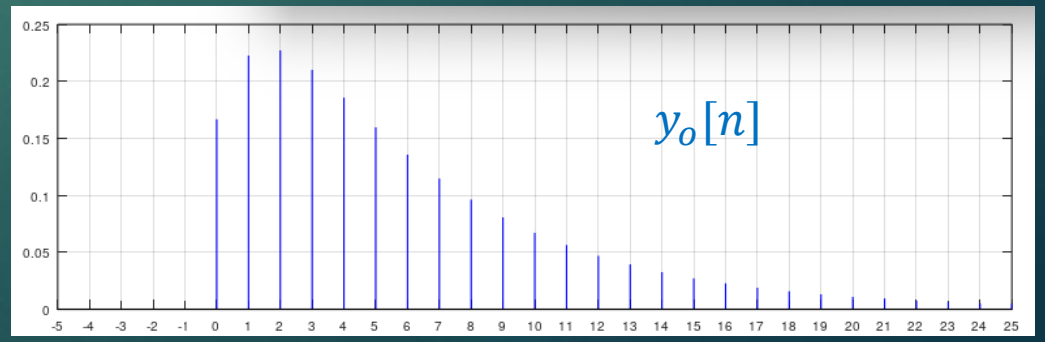



```
>> fx = @(n) (1/2).^n.*f_u(n);
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```



Resposta ao estado nulo

```
>> y_out=conv(x_in,h);
>> stem(n,y_out(1:N),'b','marker','none')
```

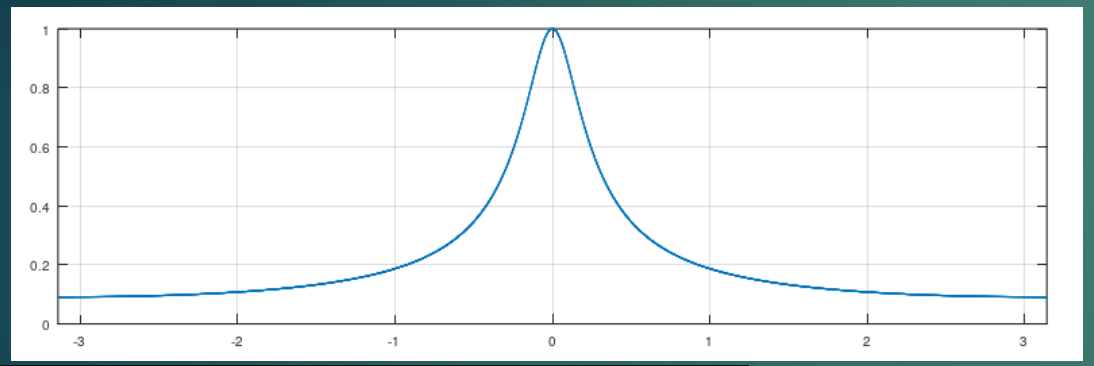


Resposta em frequência $H(jw)$ do sistema $5/6\, y[n-1] + 1/6\, x[n]$

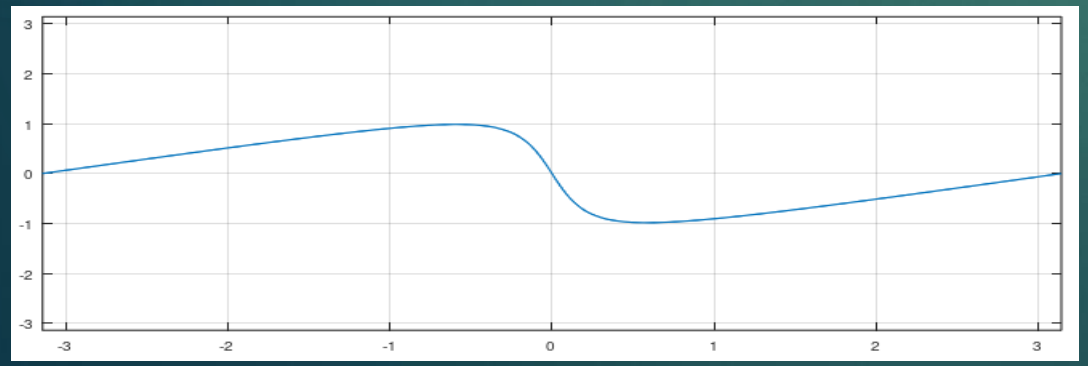
$$y[n] = \frac{5}{6}y[n-1] + \frac{1}{6}x[n] \quad \Rightarrow \quad H(z) = \frac{1/6}{1-\frac{5}{6}z^{-1}} = \frac{1/6z}{z-\frac{5}{6}}$$

$$H(jw) = H\left(z = e^{jw}\right) = \frac{1/6e^{jw}}{e^{jw}-\frac{5}{6}}, \qquad -\pi \le w \le \pi$$

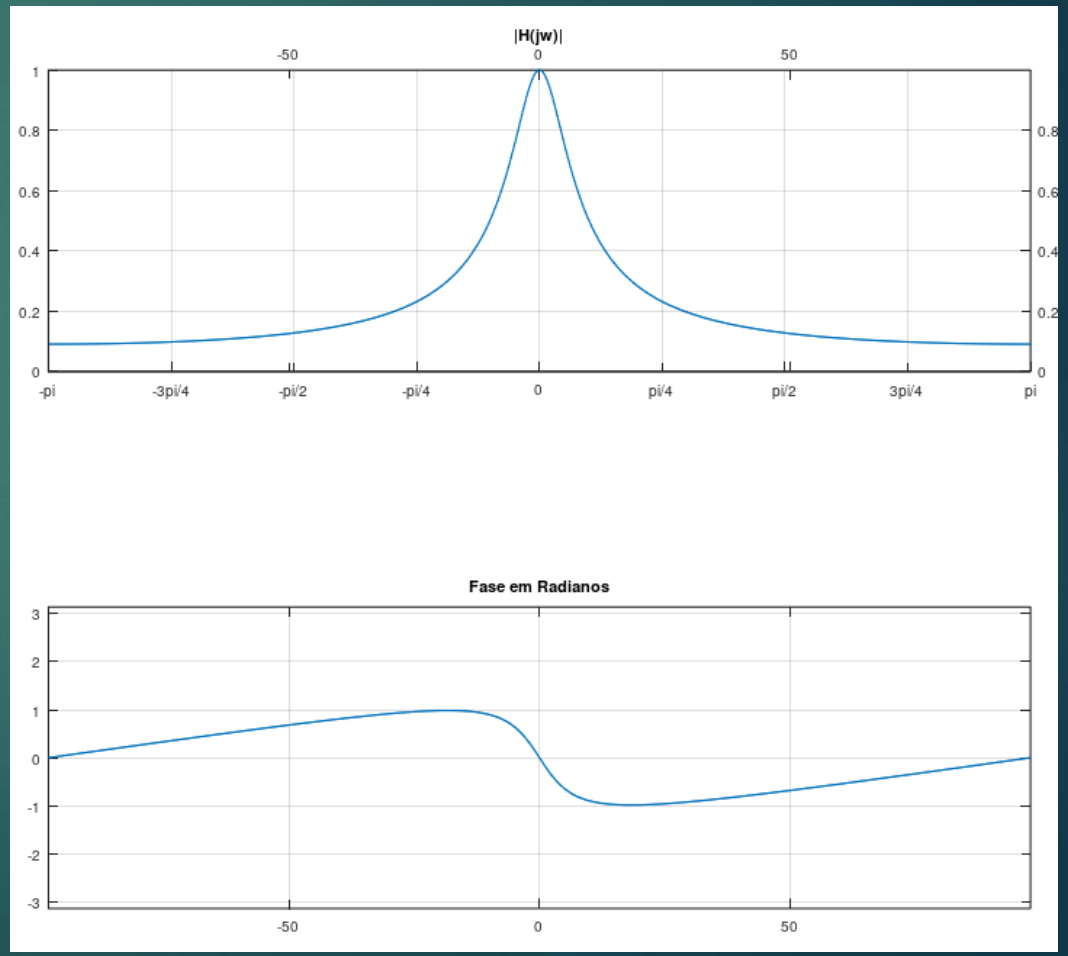
```
>> Hz=(1/6)/(1-5/6*z^-1);
>> w = linspace(-pi,pi,10000);
>> Hw=hval_fga(Hz,exp(i*w));
>> plot(w,abs(Hw))
```




```
>> plot(w,atan2(imag(Hw),real(Hw)))
```




```
>> T=0.032;
>> ws=2*pi/T
ws = 196.35
>> Hz=(1/6)/(1-5/6*z^-1);
>> Resposta_Frequencia_Hz(Hz,ws);
```



Exemplo B – Seja um sistema discreto descrito pela equação de diferenças

$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

A resposta $h(n)$ deste sistema ao impulso pode ser obtida
Analiticamente na Frequência (ou por inspeção direta)

$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

$$Y(z) = \frac{1}{3}X(z) + \frac{1}{3}z^{-1}X(z) + \frac{1}{3}z^{-2}X(z)$$

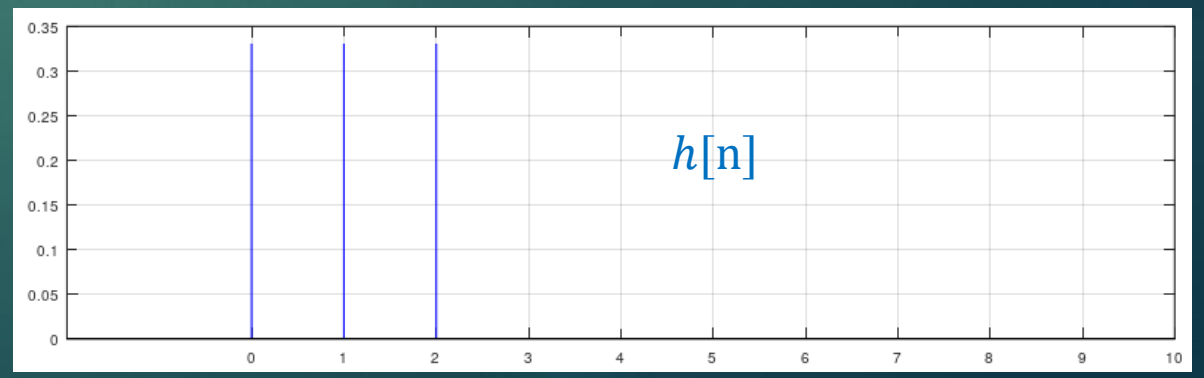
$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$$

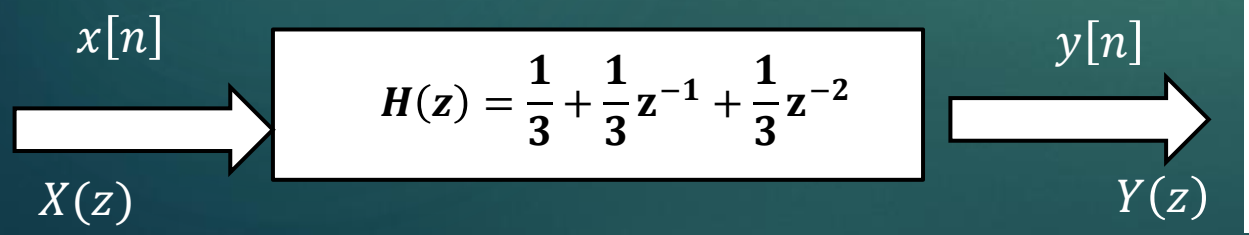
```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> Xz=1;
>> fn=itransformadaZ_Hz(Xz*Hz,1,false)
fn =

@(n) (+((0.33) * f_d (n)) + ((0.33) * f_d (n - 1)) + ((0.33) * f_d (n - 2)))
```

$$H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2} \xleftrightarrow{Z^{-1}} h(n) = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2]$$

$$h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2] = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$$

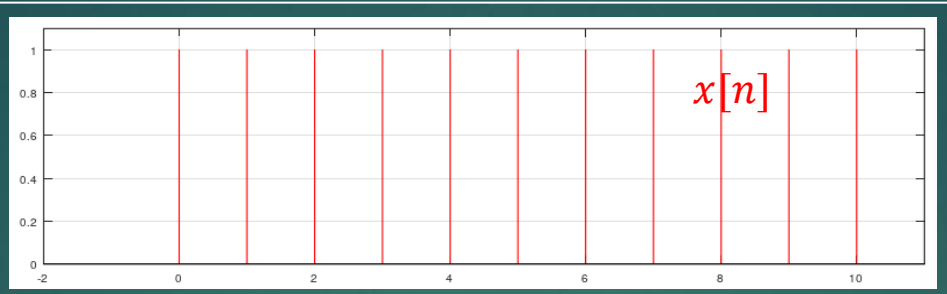
```
>> n=[-2:10];
>> h=fn(n);
>> stem(n,h,'b','marker','none')
```

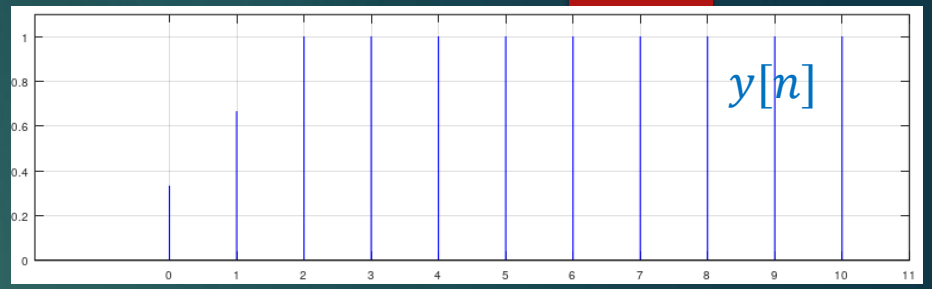




Qual a resposta do sistema $H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$ ao degrau $u[n] = \overset{Z}{\longleftrightarrow} X(z) = \frac{z}{z-1}$

```
>> Xz=z/(z-1);
>> n=[0:10];
>> x_in=f_u(n);
>> stem(n,x_in,'r','marker','none')
```





1) Numericamente pela Convolução

$$h[n] = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$$

$$x[n] = [1, 1, 1, 1, 1, 1, 1, 1 \ldots]$$

$$y[n] = x[n] * h[n] = \left[\frac{1}{3}, \frac{2}{3}, 1, 1, 1, 1, 1, 1, \ldots\right]$$

```
>> h=[1/3 1/3 1/3];
>> y_out=conv(x_in,h)
y_out =
   0.3333   0.6667   1.0000   1.0000   1.0000   1.0000   1.0000   1.0000   1.0000
   1.0000   1.0000   0.6667   0.3333

>> N=length(n);
>> stem(n,y_out(1:N),'b','marker','none')
```

2) Analiticamente na Frequência

$$H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$$

$$Y(z) = H(z)X(z) = \left(\frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}\right)\left(\frac{z}{z-1}\right) \overset{z^{-1}}{\longleftrightarrow} y[n]$$

$$Y(z) = \frac{1}{3}\left(\frac{z}{z-1} + \frac{1}{z-1} + \frac{1}{z(z-1)}\right)$$

$$y[n] = -\frac{2}{3}\delta[n] - \frac{1}{3}\delta[n-1] + u[n]$$

```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> Xz=z/(z-1);
>> fn=itransformadaZ_Hz(Hz*Xz,1,false)
fn =
@(n) (+(1.00000 * f_u (n)) + (-0.66667 * f_d (n)) + (-0.33333 * f_d (n - 1)))

>> y_out=fn(n)
y_out =
   0.3333   0.6667   1.0000   1.0000   1.0000   1.0000   1.0000   1.0000   1.0000
   1.0000   1.0000

>> stem(n,y_out,'b','marker','none')
```

3) Numericamente por Equação de Diferenças

$$H(q) = \frac{Y(q)}{X(q)} = \frac{1}{3} + \frac{1}{3}q + \frac{1}{3}q^2$$

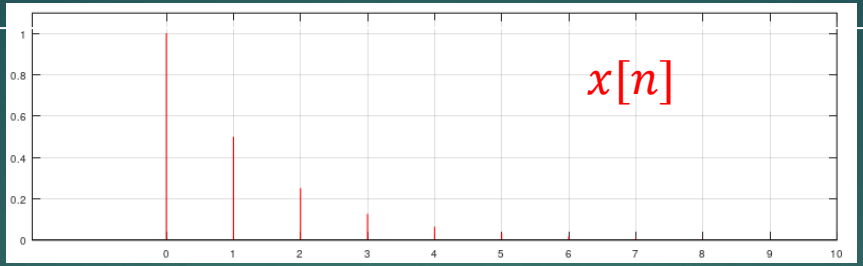
$$x[n] = [1, 1, 1, 1, 1, 1, 1, 1 \ldots]$$

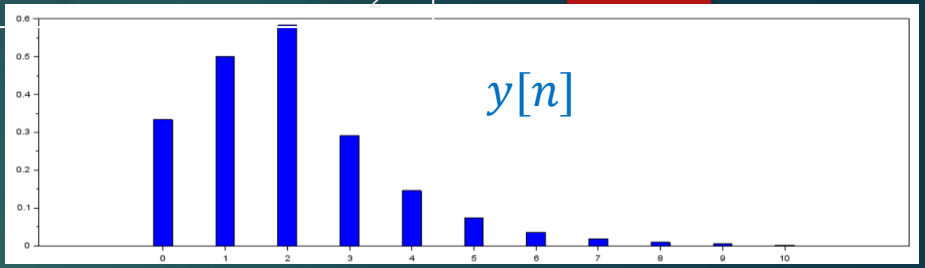
$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

```
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0])
y_out =
   0.3333   0.6667   1.0000   1.0000   1.0000   1.0000   1.0000   1.0000   1.0000
  1.0000   1.0000

>> stem(n,y_out,'b','marker','none')
```

Qual a resposta do sistema $H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$ à entrada $x[n] = \left(\frac{1}{2}\right)^n$ $x[n] \xleftrightarrow{Z} X(z) = \frac{z}{z-\frac{1}{2}}$

```
>> Xz=z/(z-1/2);
>> fx = @(n) (1/2).^n;
>> n=[0:10];
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```





1) Numericamente pela Convolução

$$h[n] = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$$

$$x[n] = \left[1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128} \cdots\right]$$

$$y[n] = x[n] * h[n] = \left[\frac{1}{3}, \frac{1}{2}, \frac{7}{12}, \frac{7}{24}, \frac{7}{48}, \frac{7}{96}, \frac{7}{192}, \frac{7}{384}\right]$$

```
>> h=[1/3 1/3 1/3];
>> y_out=conv(x_in,h)
y_out =

  0.33333     0.5         0.58333     0.29167     0.14583     0.072917     0.036458
  0.018229    0.0091146   0.0045573   0.0022786   0.00097656  0.00032552
>> stem(n,y_out(1:N),'b','marker','none')
```

2) Analiticamente na Frequência

$$H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$$

$$Y(z) = H(z)X(z) \xleftrightarrow{Z^{-1}} y(n)$$

$$y[n] = -2\delta[n] - \frac{2}{3}\delta[n-1] + \frac{7}{3}\left(\frac{1}{2}\right)^n u[n]$$

```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> fn=itransformadaZ_Hz(Hz*Xz,1,false)
fn =

@(n) (+(2.33333 * (0.50000) .^ n .* f_u (n)) + (-2.00000 * f_d (n)) + (-0.66667 * f_d (n - 1)))

>> y_out=fn(n)
y_out =

      0.33333          0.5      0.58333      0.29167      0.14583     0.072917     0.036458
0.018229    0.0091146    0.0045573    0.0022786

>> stem(n,y_out(1:N),'b','marker','none')
```

3) Numericamente por Equação de Diferenças

$$H(q) = \frac{Y(q)}{X(q)} = \frac{1}{3} + \frac{1}{3}q + \frac{1}{3}q^2$$

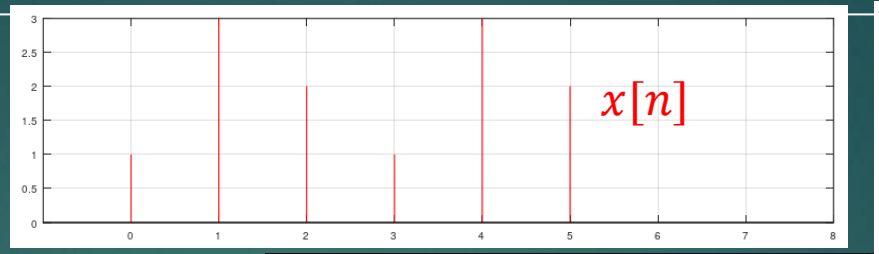
$$x[n] = \left[1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128} \cdots\right]$$

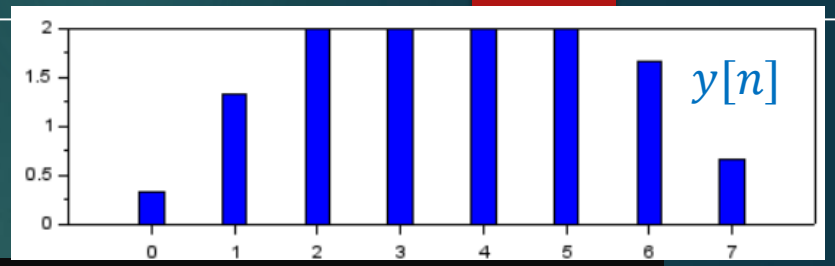
$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0])
y_out =

      0.33333          0.5      0.58333      0.29167      0.14583     0.072917     0.036458
     0.018229    0.0091146    0.0045573    0.0022786

>> stem(n,y_out(1:N),'b','marker','none')
```

Qual a resposta do sistema $\mathrm{H(z)} = \frac{1}{3} + \frac{1}{3}\mathrm{z}^{-1} + \frac{1}{3}\mathrm{z}^{-2}$ à entrada $x[n] = [1,3,2,1,3,2,0,0] \xleftrightarrow{Z} X(z) = 1 + 3z^{-1} + 2z^{-2} + z^{-3} + 3z^{-4} + 2z^{-5}$

```
>> x_in=[1,3,2,1,3,2,0,0];
>> Xz=minreal(1+3*z^-1+2*z^-2+z^-3+3*z^-4+2*z^-5)

      z^5 + 3 z^4 + 2 z^3 + 1 z^2 + 3 z + 2
 y1:  -------------------------------------
                          z^5

>> n=[0:7];
>> stem(n,x_in,'r','marker','none')
```





1) Numericamente pela Convolução

$$h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2] = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right]$$

$$x[n] = [1,3,2,1,3,2,0,0]$$

$$y[n] = x[n] * h[n] = \left[\frac{1}{3}\ \frac{4}{3}\ 2\ 2\ 2\ 2\ \frac{5}{3}\ \frac{2}{3}\right]$$

```
>> h=[1/3 1/3 1/3];
>> N=length(n);
>> y_out=conv(x_in,h)
y_out =

      0.33333       1.3333            2            2            2            2
       1.6667      0.66667            0            0

>> stem(n,y_out(1:N),'b','marker','none')
```

2) Analiticamente na Frequência

$$\mathrm{H(z)} = \frac{1}{3} + \frac{1}{3}\mathrm{z}^{-1} + \frac{1}{3}\mathrm{z}^{-2}$$

$$Y(z) = H(z)X(z) \xleftrightarrow{Z^{-1}} y(n)$$

$$y[n] = \frac{1}{3}\delta[n] + \frac{4}{3}\delta[n-1] + 2\delta[n-2] + 2\delta[n-3] + 2\delta[n-4] + 2\delta[n-5] + \frac{5}{3}\delta[n-6] + \frac{2}{3}\delta[n-7] = \left[\frac{1}{3}\ \frac{4}{3}\ 2\ 2\ 2\ 2\ \frac{5}{3}\ \frac{2}{3}\right]$$

```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> fn=itransformadaZ_Hz(Hz*Xz,1,false)
fn =

@(n) (+((0.33) * f_d (n)) + ((1.33) * f_d (n - 1)) + ((2.00) * f_d (n - 2)) +
((2.00) * f_d (n - 3)) + ((2.00) * f_d (n - 4)) + ((2.00) * f_d (n - 5)) + ((1
.67) * f_d (n - 6)) + ((0.67) * f_d (n - 7)))

>> y_out=fn(n)
y_out =

         0.33         1.33            2            2            2            2
         1.67         0.67

>> stem(n,y_out,'b','marker','none')
```

3) Numericamente por Equação de Diferenças

$$\mathrm{H(q)} = \frac{\mathrm{Y(q)}}{\mathrm{X(q)}} = \frac{1}{3} + \frac{1}{3}q + \frac{1}{3}q^2$$

$$x[n] = [1,3,2,1,3,2,0,0]$$

$$y[n] = \frac{1}{3}\mathrm{x}[n] + \frac{1}{3}\mathrm{x}[n-1] + \frac{1}{3}\mathrm{x}[n-2]$$

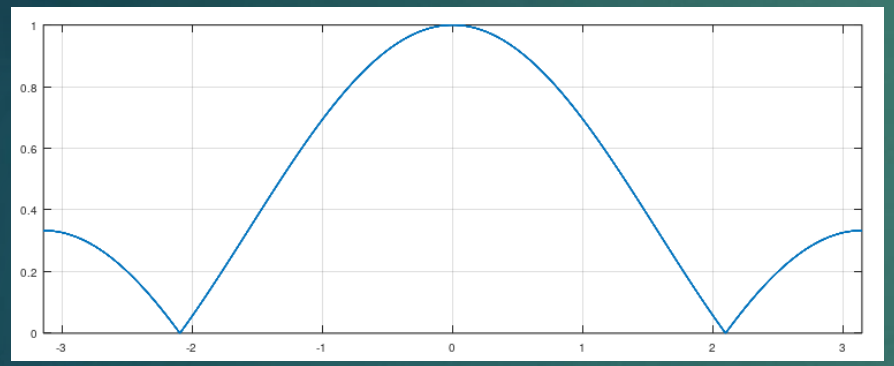
```
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0])
y_out =

      0.33333       1.3333            2            2            2            2
      1.6667      0.66667

>> stem(n,y_out,'b','marker','none')
```

Calcule resposta em frequência $H(jw)$ do sistema $H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$

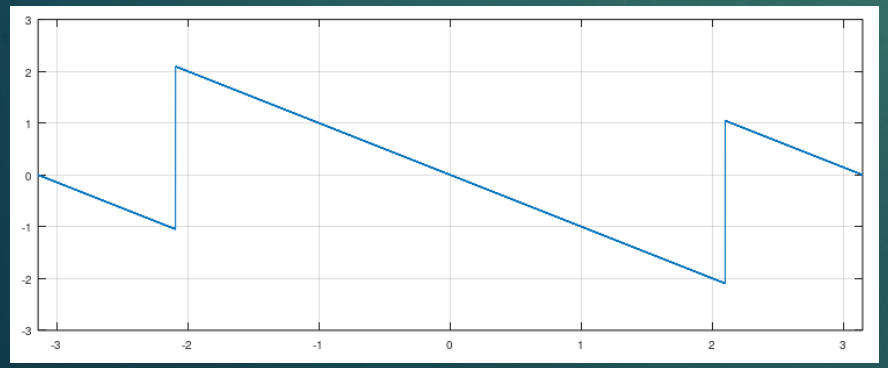
$$H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$$

$$H(jw) = H(z = e^{jw}) = \frac{1}{3} + \frac{1}{3}(e^{jw})^{-1} + \frac{1}{3}(e^{jw})^{-2}; \quad -\pi \leq w \leq \pi$$

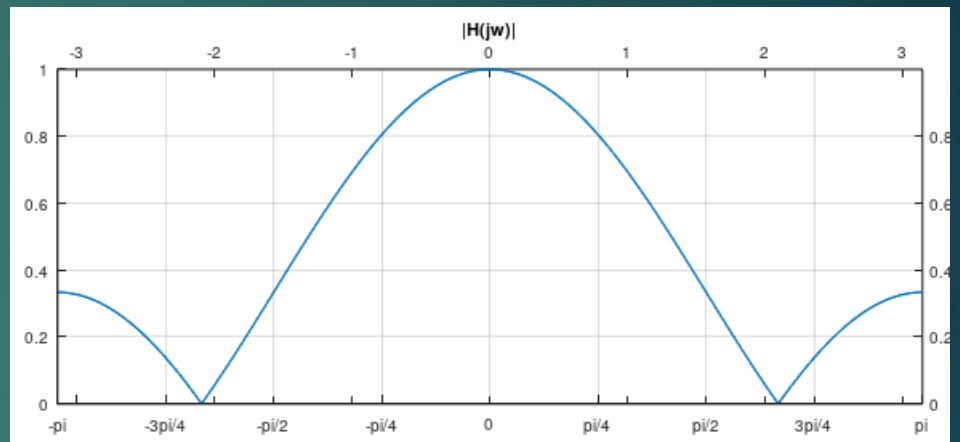
```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> w = linspace(-pi,pi,10000);
>> Hw=hval_fga(Hz,exp(i*w));
>> plot(w,abs(Hw))
```

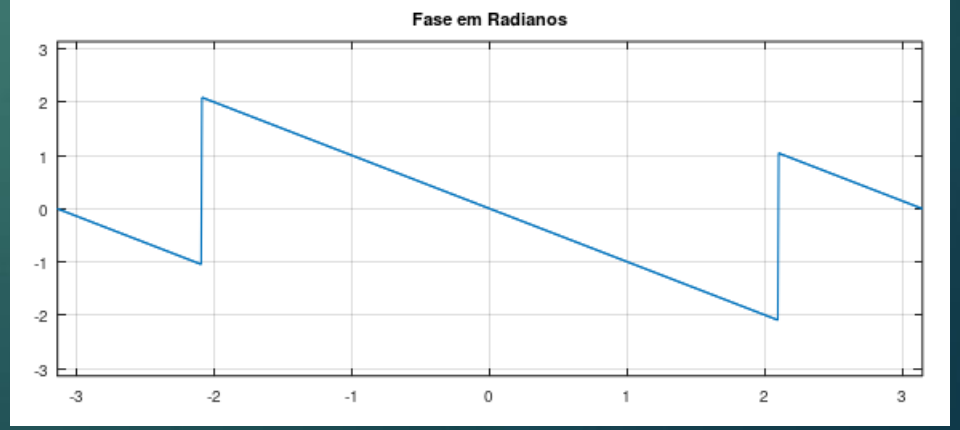


```
>> plot(w,atan2(imag(Hw),real(Hw)))
```



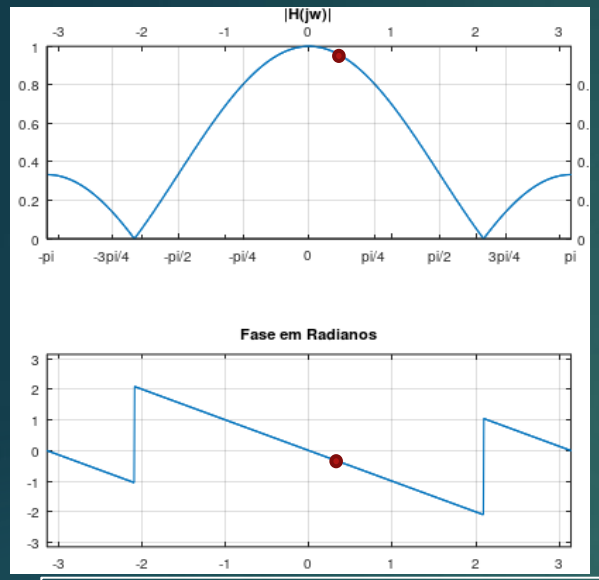
```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> Resposta_Frequencia_Hz(Hz,2*pi);
```





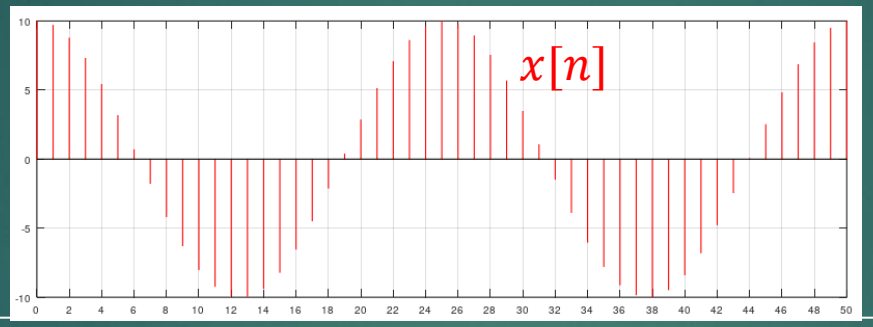
Qual a resposta do sistema $H(q) = \frac{1}{3} + \frac{1}{3}q + \frac{1}{3}q^2$ à entrada $x[n] = 10\,cos\left(\frac{1}{4}n\right)$

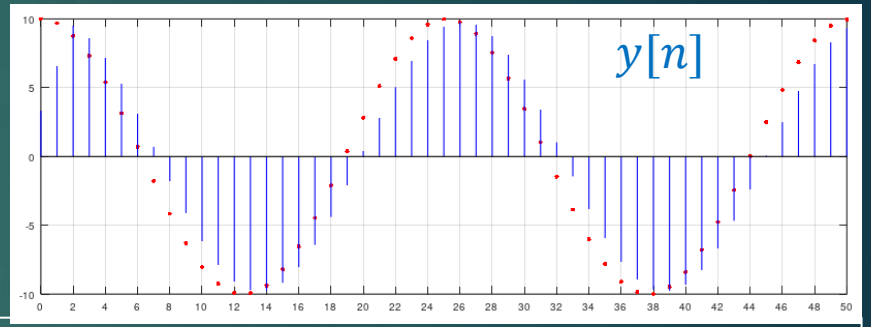
Numericamente por Equação de Diferenças



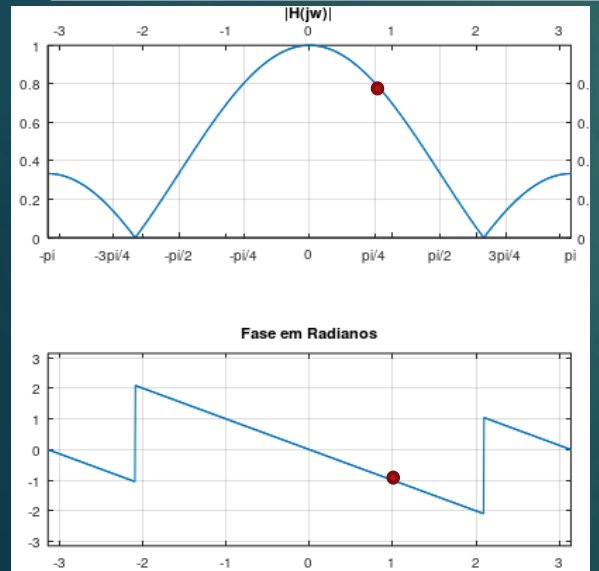

```
>> fx = @(n) 10*cos(1/4*n);
>> n=[0:50];
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```

```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0]);
>> grid on; hold on;
>> stem(n,y_out,'b','marker','none')
>> plot(n,x_in,'r.')
```



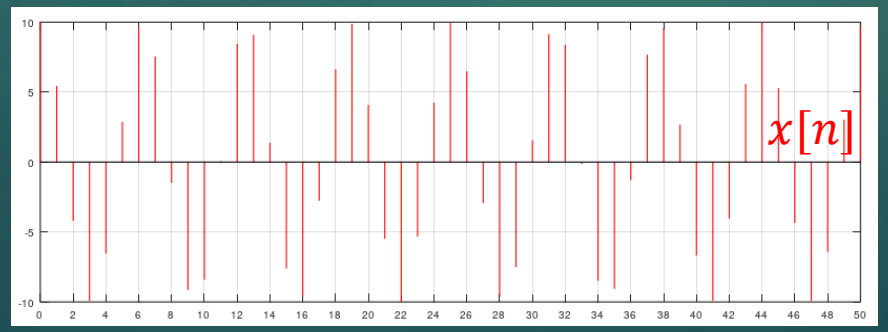


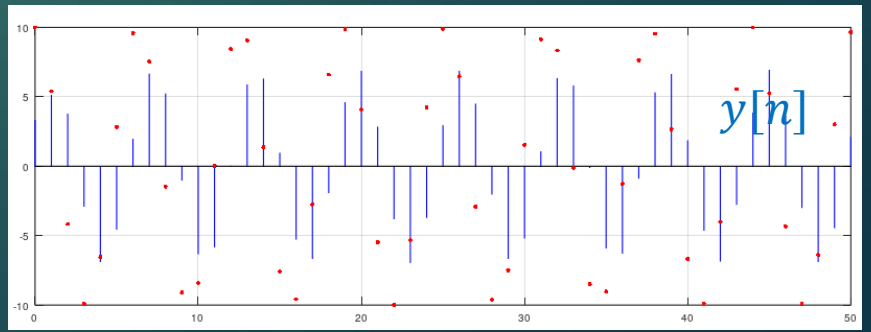
Qual a resposta do sistema $H(z) = \frac{1}{3} + \frac{1}{3}z^{-1} + \frac{1}{3}z^{-2}$ à entrada $x[n] = 10\,cos(1.0\,n)$




```
>> fx = @(n) 10*cos(1*n);
>> n=[0:50];
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```

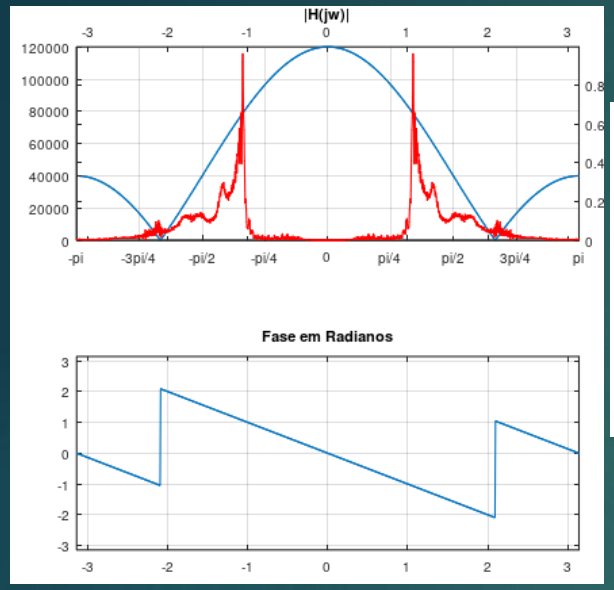
```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0]);
>> grid on; hold on;
>> stem(n,y_out,'b','marker','none')
>> plot(n,x_in,'r.')
```



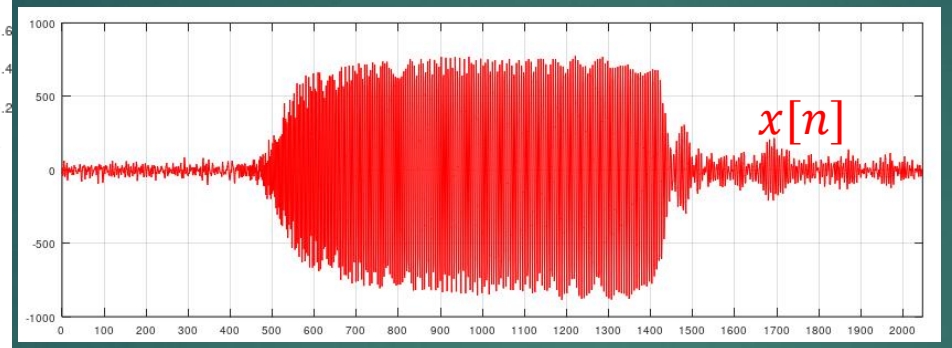


Qual a resposta do sistema $H(q) = \frac{1}{3} + \frac{1}{3}q + \frac{1}{3}q^2$ à entrada $x[n] = som\ do\ morcego$

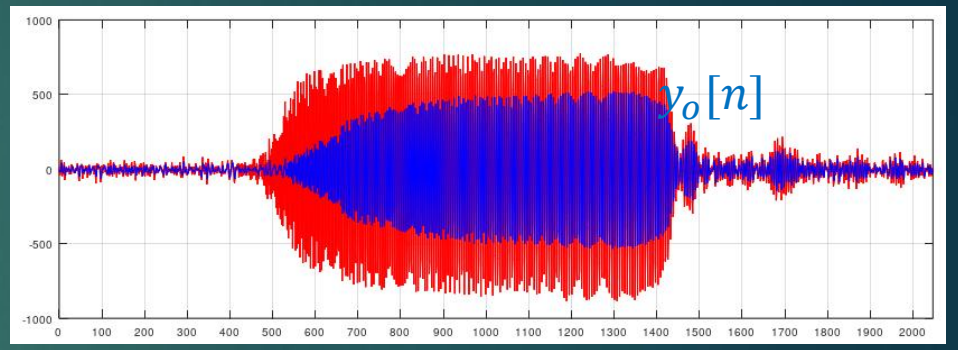
Numericamente por Equação de Diferenças

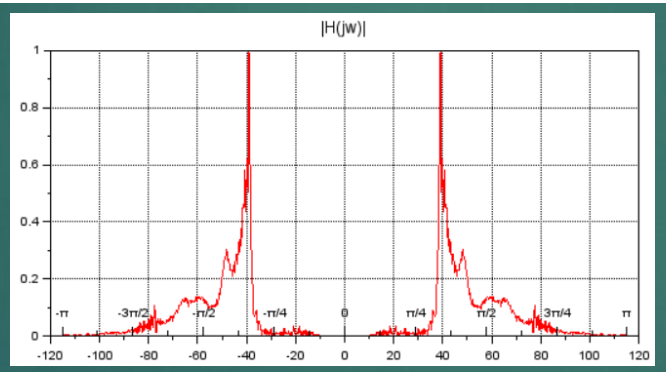


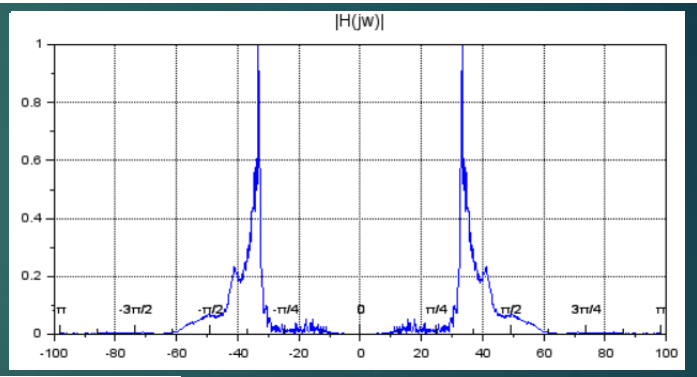

```
>> load('./Dados/bat.mat')
>> x_in=bat';
>> n=[0:length(x_in)-1];
>> plot(n,x_in,'r')
```

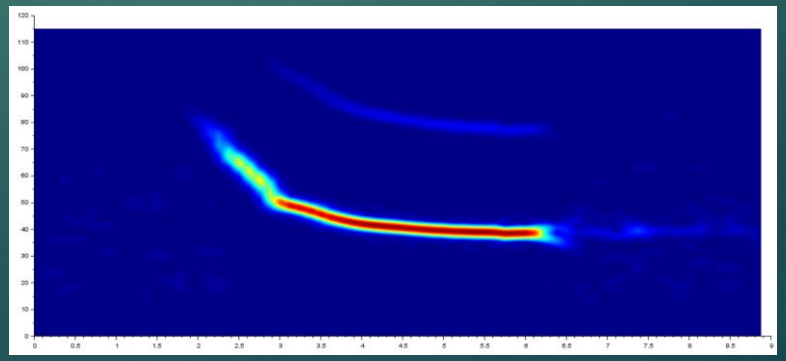



```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0]);
>> hold on; grid on
>> plot(n,x_in,'r')
>> plot(n,y_out,'b')
```







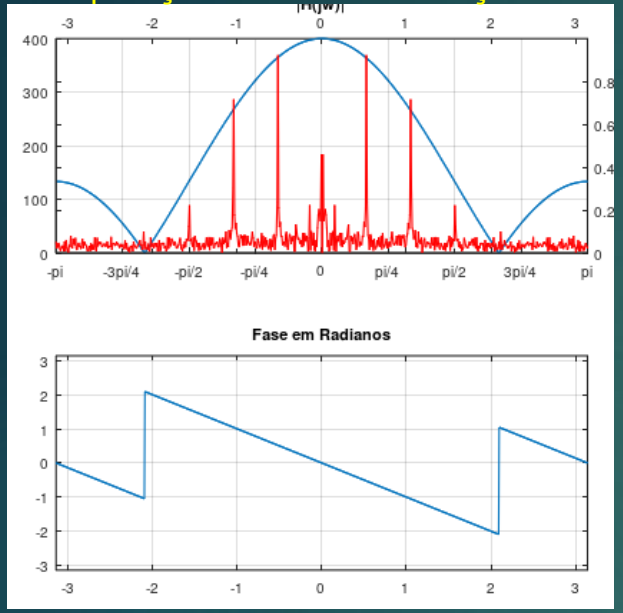


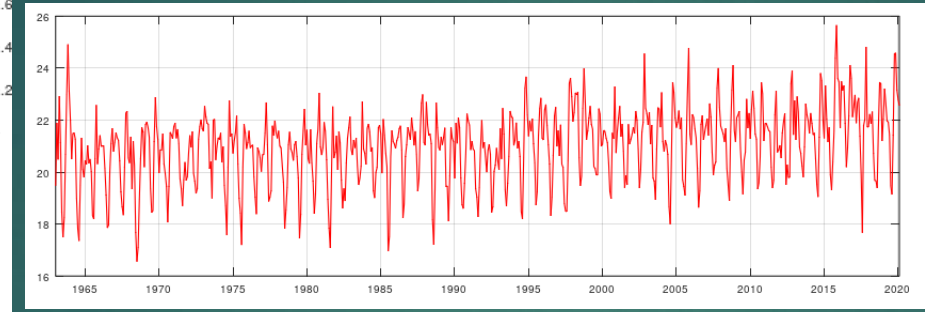
Qual a resposta do sistema $H(q) = \frac{1}{3} + \frac{1}{3}q + \frac{1}{3}q^2$ à entrada $x[n] = temperatura\ DF$

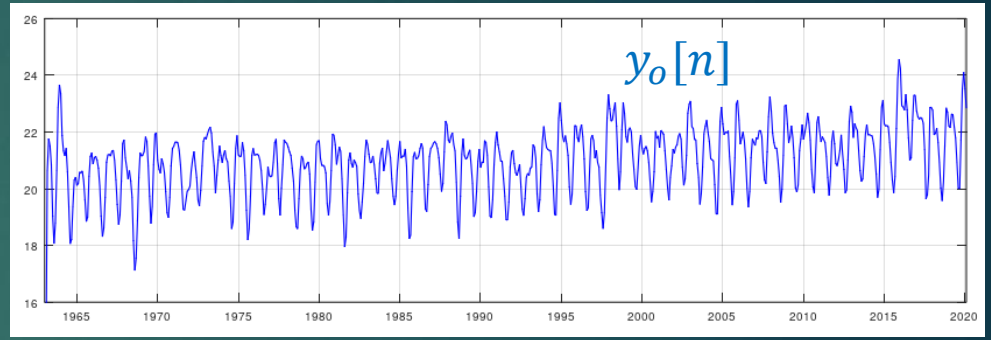
Numericamente por Equação de Diferenças

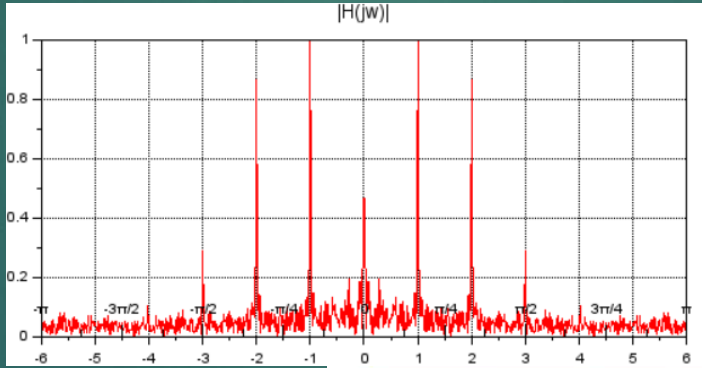
```
>> [n,x_in]=textread("./Dados/temperatura_DF_2.txt",
"%f %f",686);
>>
>> plot(n,x_in,'r')
```

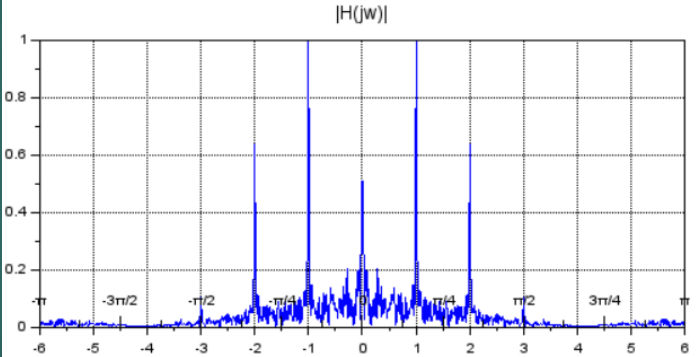
```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0]);
>> plot(n,y_out,'b')
```

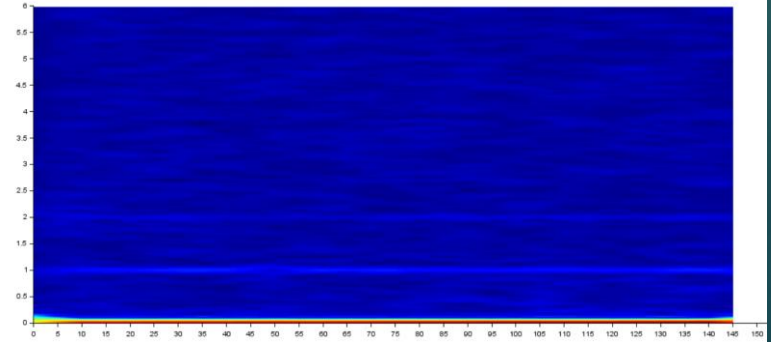












Exemplo C – Seja um sistema discreto descrito pela equação de diferenças

$$y[n] = \left(\frac{57}{40}\right) y[n-1] - \left(\frac{37}{33}\right) y[n-2] + \left(\frac{6}{19}\right) y[n-3] + \left(\frac{5}{49}\right) x[n] + \left(\frac{3}{34}\right) x[n-1] + \left(\frac{3}{34}\right) x[n-2] + \left(\frac{5}{49}\right) x[n-3]$$

A Função de Transferência $H(z)$ pode ser obtida diretamente da Equação de Diferenças

$$y[n] = \left(\frac{57}{40}\right) y[n-1] - \left(\frac{37}{33}\right) y[n-2] + \left(\frac{37}{33}\right) y[n-3] + \left(\frac{5}{49}\right) x[n] + \left(\frac{3}{34}\right) x[n-1] + \left(\frac{3}{34}\right) x[n-2] + \left(\frac{5}{49}\right) x[n-3]$$

$$\mathrm{H(z)} = \frac{\left(\frac{5}{49}\right) + \left(\frac{3}{34}\right) z^{-1} + \left(\frac{3}{34}\right) z^{-2} + \left(\frac{5}{49}\right) z^{-3}}{1 - \left(\frac{57}{40}\right) z^{-1} + \left(\frac{37}{33}\right) z^{-2} - \left(\frac{6}{19}\right) z^{-3}}$$

```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> fn=itransformadaZ_Hz(Xz*Hz,1,false)
fn =

@(n) (+(2 * (0.20504) * (0.82012) .^ n .* ((-0.99061) * cos (n * (0.94896)) + (0.13672) * sin
(n * (0.94896))) .* f_u (n)) + (0.83139 * (0.46950) .^ n .* f_u (n)) + (-0.32313 * f_d (n)))

>> n=[-5:25];
>> stem(n,fn(n),'b','marker','none')
```

A resposta $h(n)$ deste sistema ao impulso pode ser obtida Analiticamente na Frequência

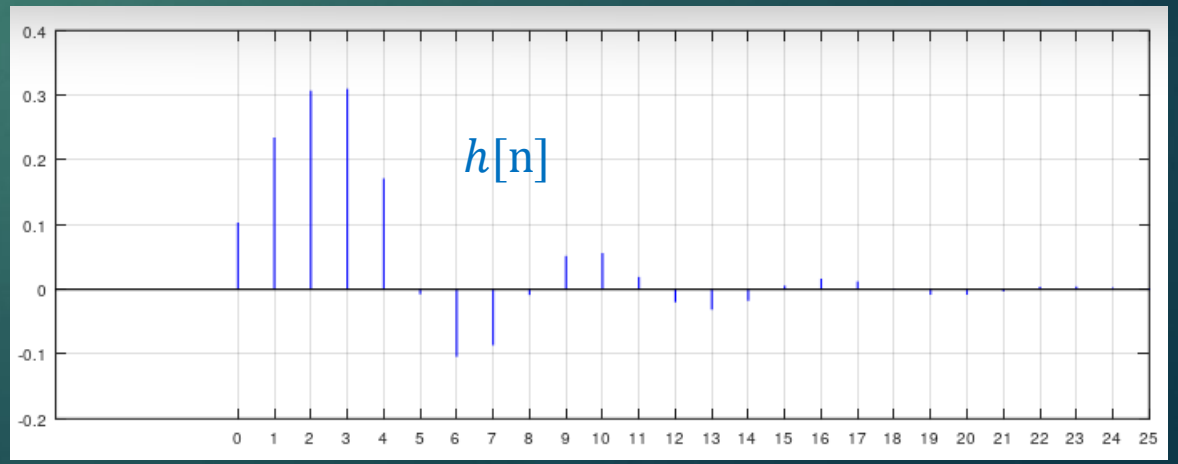
$$H(z) = \frac{\left(\frac{5}{49}\right) z^3 + \left(\frac{3}{34}\right) z^2 + \left(\frac{3}{34}\right) z + \left(\frac{5}{49}\right)}{z^3 - \left(\frac{57}{40}\right) z^2 + \left(\frac{37}{33}\right) z - \left(\frac{6}{19}\right)}$$

$$H(z) \xleftrightarrow{z^{-1}} h[n]$$

$x[n]$ → $X(z)$

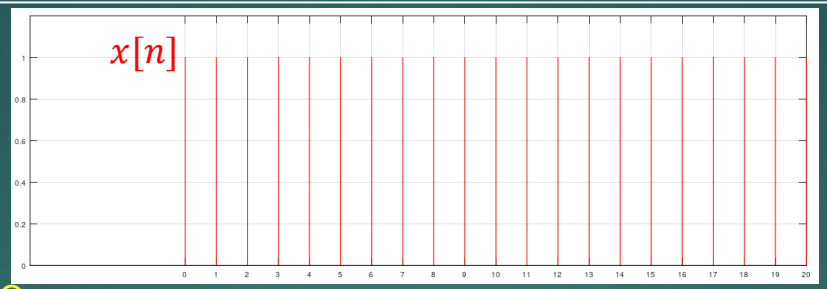
$$H(z) = \frac{\left(\frac{5}{49}\right) z^3 + \left(\frac{3}{34}\right) z^2 + \left(\frac{3}{34}\right) z + \left(\frac{5}{49}\right)}{z^3 - \left(\frac{57}{40}\right) z^2 + \left(\frac{37}{33}\right) z - \left(\frac{6}{19}\right)}$$

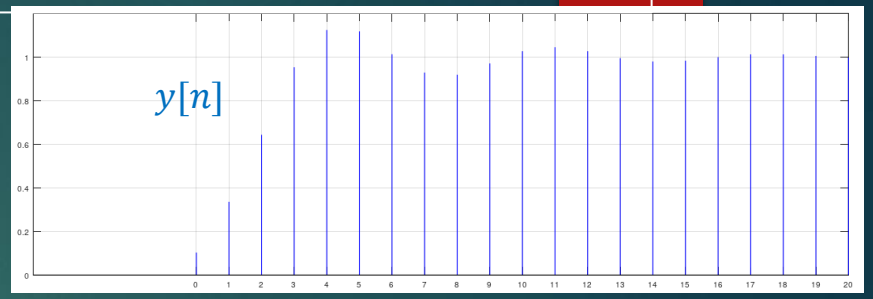
→ $y[n]$ / $Y(z)$



Qual a resposta do sistema $H(q) = \frac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ ao degrau $u[n] = \xleftrightarrow{Z} X(z) = \frac{z}{z-1}$

```
>> Xz=z/(z-1);
>> N=21;
>> n=[0:N-1];
>> x_in=f_u(n);
>> stem(n,x_in,'r','marker','none')
```





1) Numericamente pela Convolução

Como $h(n)$ tem duração infinita, a Convolução no tempo não pode a principio ser feita. A não ser que $h(n)$ seja truncada. Neste caso teremos um erro de truncamento. O resultado será aproximado.

```
>> y_out=conv(x_in,h);   ??????????
>> stem(n,y_out(1:N),'b','marker','none')
```

2) Analiticamente na Frequência

$$H(z) = \frac{\left(\frac{5}{49}\right)z^3 + \left(\frac{3}{34}\right)z^2 + \left(\frac{3}{34}\right)z + \left(\frac{5}{49}\right)}{z^3 - \left(\frac{57}{40}\right)z^2 + \left(\frac{37}{33}\right)z - \left(\frac{6}{19}\right)}$$

$$Y(z) = H(z)X(z) \xleftrightarrow{z^{-1}} y[n]$$

$$y[n] = f(n)$$

```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> fn=itransformadaZ_Hz(Hz*z/(z-1),1,false)
fn =

@(n) (+(1.00034 * f_u (n)) + (2 * (0.19857) * (0.82012) .^ n .* ((-0.40917)
 * cos (n * (0.94896)) + (-0.91246) * sin (n * (0.94896))) .* f_u (n)) + (-
0.73580 * (0.46950) .^ n .* f_u (n)))

>> stem(n,fn(n),'b','marker','none')
```

3) Numericamente por Equação de Diferenças

$$H(z^{-1}) = \frac{\left(\frac{5}{49}\right) + \left(\frac{3}{34}\right)z^{-1} + \left(\frac{3}{34}\right)z^{-2} + \left(\frac{5}{49}\right)z^{-3}}{1 - \left(\frac{57}{40}\right)z^{-1} + \left(\frac{37}{33}\right)z^{-2} - \left(\frac{6}{19}\right)z^{-3}}$$

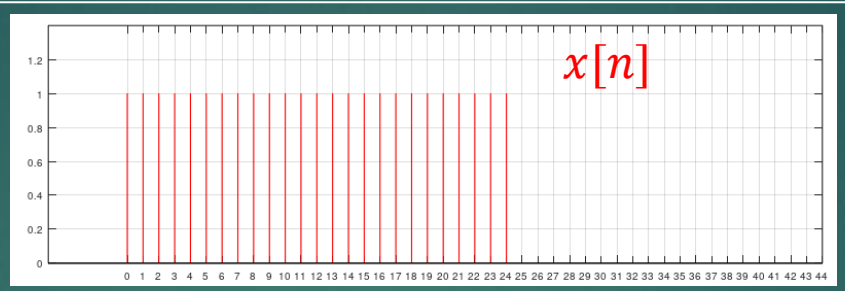
$$y[n] = \left(\frac{57}{40}\right)y[n-1] - \left(\frac{37}{33}\right)y[n-2] + \left(\frac{37}{33}\right)y[n-3] + \left(\frac{5}{49}\right)x[n] + \left(\frac{3}{34}\right)x[n-1] + \left(\frac{3}{34}\right)x[n-2] + \left(\frac{5}{49}\right)x[n-3]$$

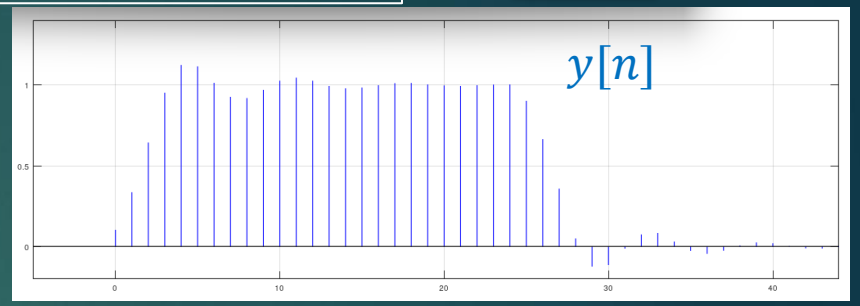
```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> y_out=applyIIR(Hz,x_in,[0 0 0 0],[0 0 0 0]);
>> stem(n,fn(n),'b','marker','none')
```

Qual a resposta do sistema $H(q) = \frac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ um pulso de largura 25

$$x[n] = u[n] - u[n-25] \xleftrightarrow{Z} X(z) = \frac{z}{z-1} - z^{-25}\frac{z}{z-1} = \frac{z - z^{-24}}{z-1}$$

```
>> Xz=(z-z^-24)/(z-1);
>> N=45;
>> n=[0:N-1];
>> x_in=f_u(n)-f_u(n-25);
>> stem(n,x_in,'r','marker','none')
```





1) Numericamente pela Convolução

Truncando h(n), o resultado será aproximado.

```
>> y_out=conv(x_in,h);   ??????????
>> stem(n,y_out(1:N),'b','marker','none')
```

2) Analiticamente na Frequência

$$H(z) = \frac{\left(\frac{5}{49}\right)z^3 + \left(\frac{3}{34}\right)z^2 + \left(\frac{3}{34}\right)z + \left(\frac{5}{49}\right)}{z^3 - \left(\frac{57}{40}\right)z^2 + \left(\frac{37}{33}\right)z - \left(\frac{6}{19}\right)}$$

$$Y(z) = H(z)X(z) \xleftrightarrow{Z^{-1}} y[n]$$

$$y_u[n] = f(n)$$

```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> fn=itransformadaZ_Hz(Hz*z/(z-1),1,false)
fn =

@(n) (+(1.00034 * f_u (n)) + (2 * (0.19857) * (0.82012) .^ n .* ((-0.40917)
 * cos (n * (0.94896)) + (-0.91246) * sin (n * (0.94896))) .* f_u (n)) + (-
0.73580 * (0.46950) .^ n .* f_u (n)))

>> y_out=fn(n)-fn(n-25);
>> stem(n,y_out,'b','marker','none')
```

3) Numericamente por Equação de Diferenças

$$H(q) = \frac{\left(\frac{5}{49}\right) + \left(\frac{3}{34}\right)q + \left(\frac{3}{34}\right)q^2 + \left(\frac{5}{49}\right)q^3}{1 - \left(\frac{57}{40}\right)q + \left(\frac{37}{33}\right)q^2 - \left(\frac{6}{19}\right)q^3}$$

```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> y_out=applyIIR(Hz,x_in,[0 0 0 0],[0 0 0 0]);
>> stem(n,y_out,'b','marker','none')
```

$$y[n] = \left(\frac{57}{40}\right)y[n-1] - \left(\frac{37}{33}\right)y[n-2] + \left(\frac{37}{33}\right)y[n-3] + \left(\frac{5}{49}\right)x[n] + \left(\frac{3}{34}\right)x[n-1] + \left(\frac{3}{34}\right)x[n-2] + \left(\frac{5}{49}\right)x[n-3]$$

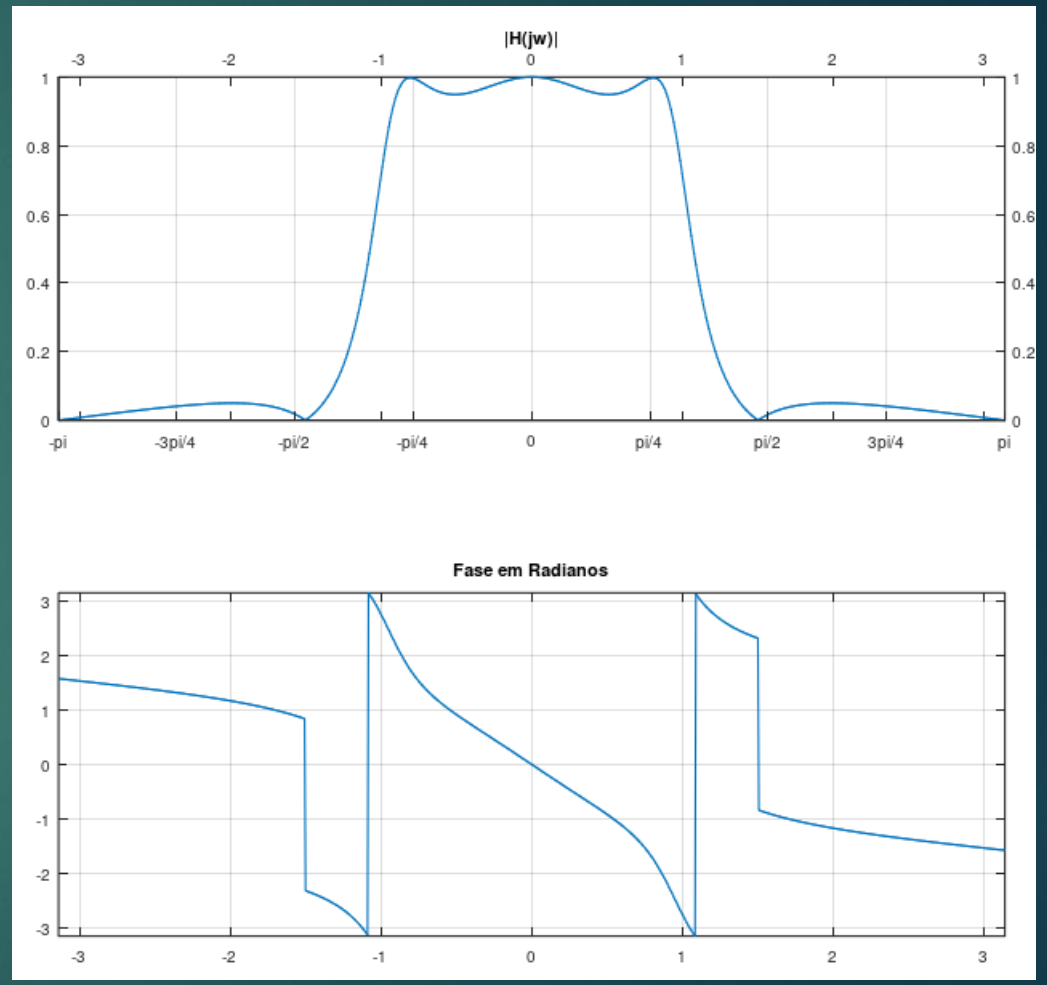
Calcule resposta em frequência $H(jw)$ do sistema $\mathrm{H(z)} = \dfrac{\left(\frac{5}{49}\right)z^3+\left(\frac{3}{34}\right)z^2+\left(\frac{3}{34}\right)z+\left(\frac{5}{49}\right)}{z^3-\left(\frac{57}{40}\right)z^2+\left(\frac{37}{33}\right)z-\left(\frac{6}{19}\right)}$

$$\mathrm{H(z)} = \frac{\left(\frac{5}{49}\right)z^3+\left(\frac{3}{34}\right)z^2+\left(\frac{3}{34}\right)z+\left(\frac{5}{49}\right)}{z^3-\left(\frac{57}{40}\right)z^2+\left(\frac{37}{33}\right)z-\left(\frac{6}{19}\right)}$$

$$H(jw) = H\left(z = e^{-jw}\right) = \frac{\left(\frac{5}{49}\right)e^{j3w}+\left(\frac{3}{34}\right)e^{j2w}+\left(\frac{3}{34}\right)e^{jw}+\left(\frac{5}{49}\right)}{e^{j3w}-\left(\frac{57}{40}\right)e^{j2w}+\left(\frac{37}{33}\right)e^{jw}-\left(\frac{6}{19}\right)}$$

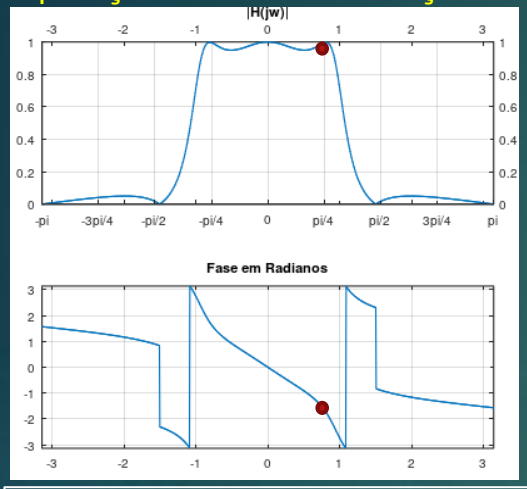
$$-\pi \leq w \leq \pi$$

```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> Resposta_Frequencia_Hz(Hz,2*pi);
```



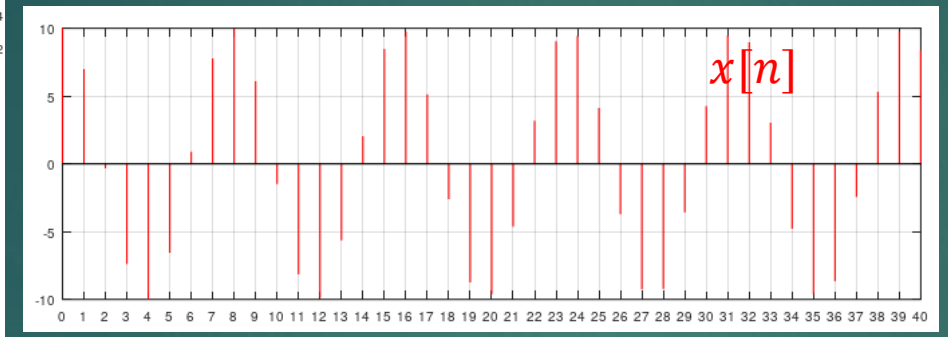
Qual a resposta do sistema $H(q) = \dfrac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ à entrada $x[n] = 10\,cos(0.8\,n)$

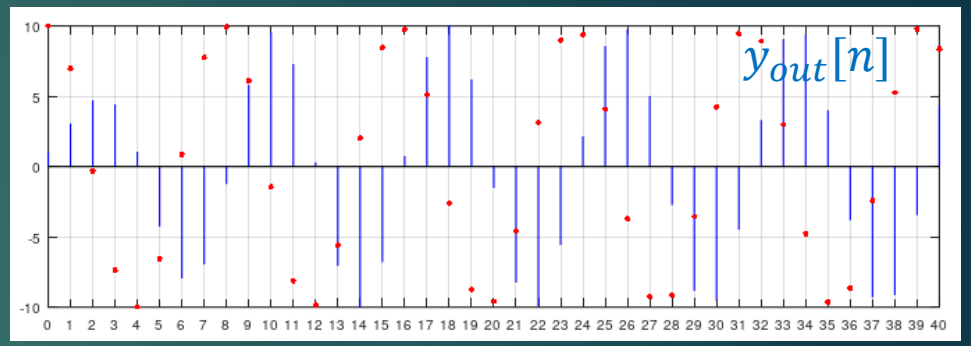
Numericamente por Equação de Diferenças



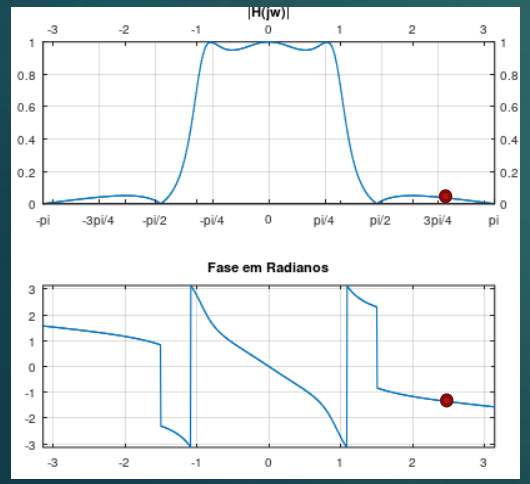

```
>> fx = @(n) 10*cos(0.8*n);
>> n=[0:40];
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```

```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> y_out=applyIIR(Hz,x_in,[0 0 0 0],[0 0 0 0]);
>> stem(n,y_out,'b','marker','none')
>> plot(n,x_in,'r.')
```



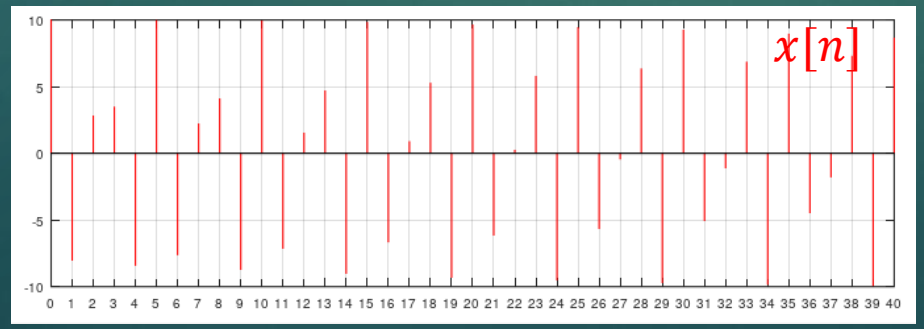


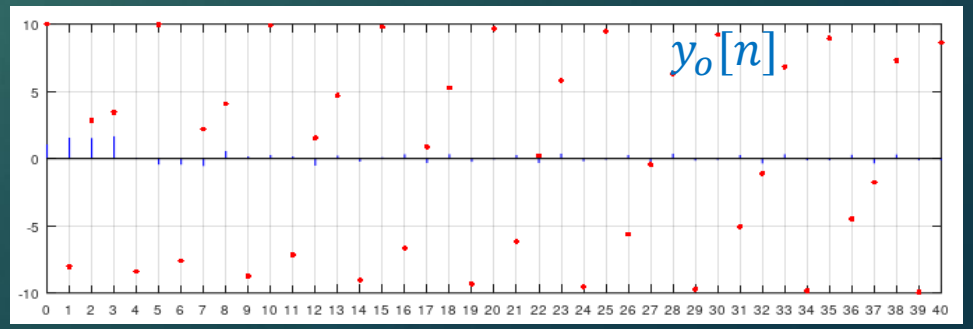
Qual a resposta do sistema $H(q) = \dfrac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ à entrada $x[n] = 10\,cos(2.5n)$




```
>> fx = @(n) 10*cos(2.5*n);
>> n=[0:40];
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```

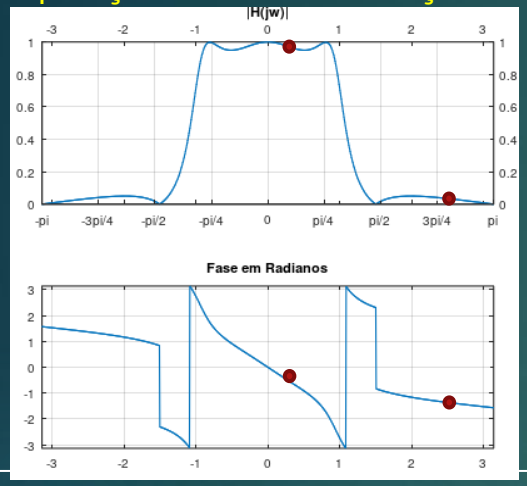
```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> y_out=applyIIR(Hz,x_in,[0 0 0 0],[0 0 0 0]);
>> stem(n,y_out,'b','marker','none')
>> plot(n,x_in,'r.')
```





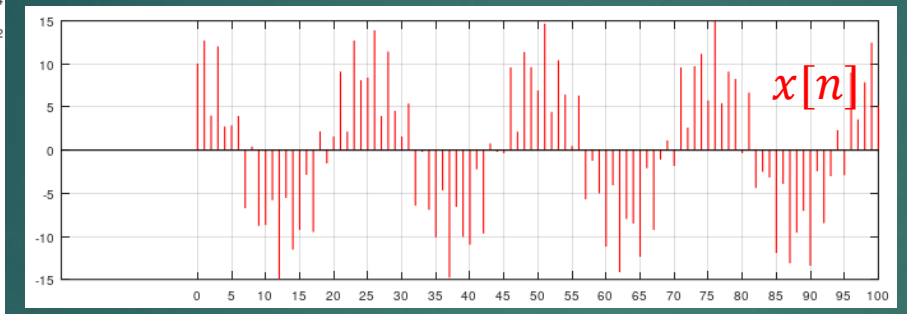
Qual a resposta do sistema $H(q) = \frac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ à entrada $x[n] = 10\cos\left(\frac{1}{4}n\right) + 5\sin(2.5n)$

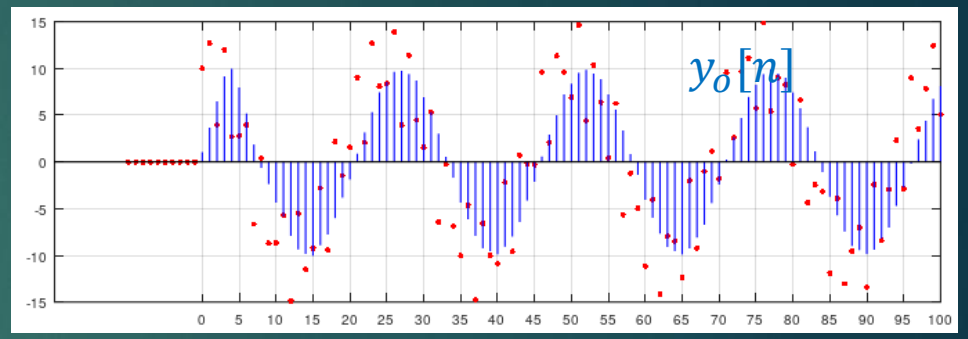
Numericamente por Equação de Diferenças



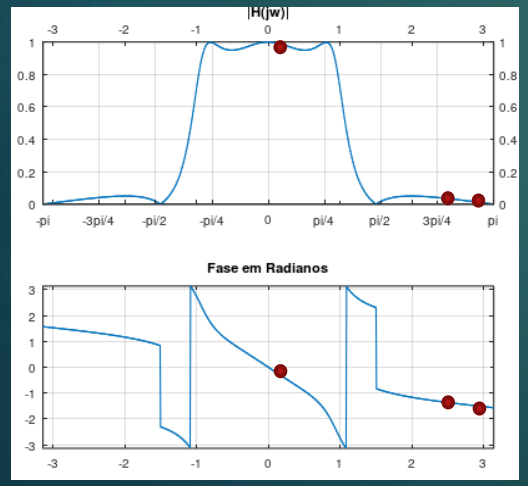

```
>> fx = @(n) (10*cos(1/4*n)+5*sin(2.5*n)).*f_u(n);
>> n=[-10:100];
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```

```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> y_out=applyIIR(Hz,x_in,[0 0 0 0],[0 0 0 0]);
>> stem(n,y_out,'b','marker','none')
>> plot(n,x_in,'r.')
```



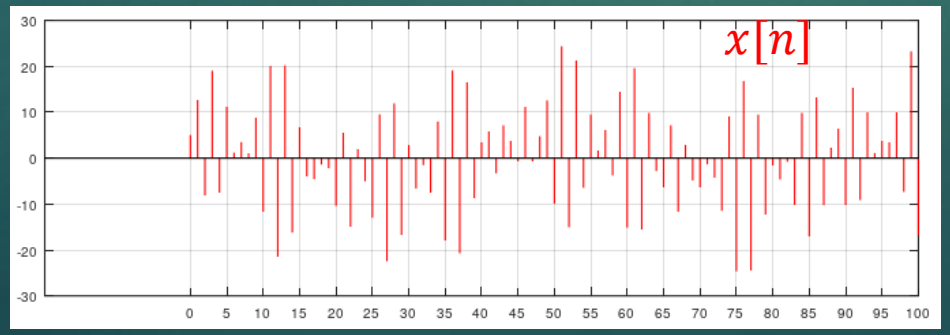


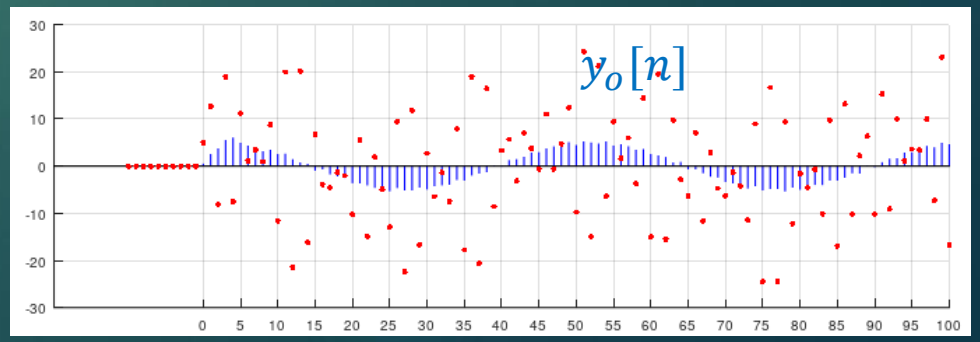
Qual a resposta do sistema $H(q) = \frac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ à entrada $x[n] = 5\cos\left(\frac{1}{8}n\right) + 10\sin(2.5n) + 12\sin(3n)$




```
>> fx = @(n) (5*cos(1/8*n)+10*sin(2.5*n)+12*sin(3*n)).*f_u(n);
>> n=[-10:100];
>> x_in=fx(n);
>> stem(n,x_in,'r','marker','none')
```

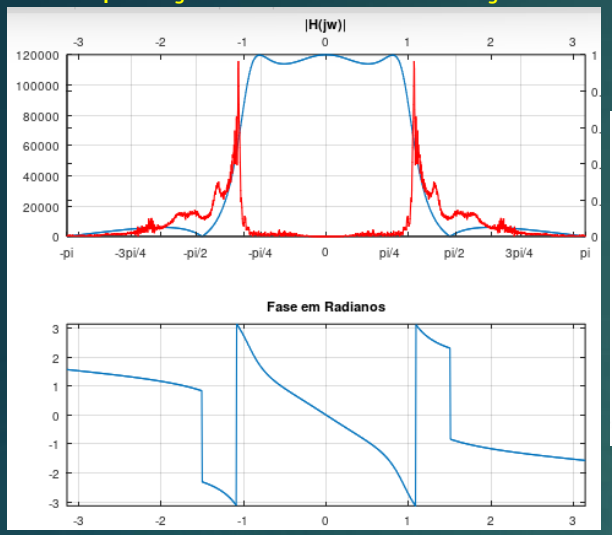
```
>> Hz=(5/49*z^3+3/34*z^2+3/34*z+5/49)/(z^3-57/40*z^2+37/33*z-6/19);
>> y_out=applyIIR(Hz,x_in,[0 0 0 0],[0 0 0 0]);
>> stem(n,y_out,'b','marker','none')
>> plot(n,x_in,'r.')
```



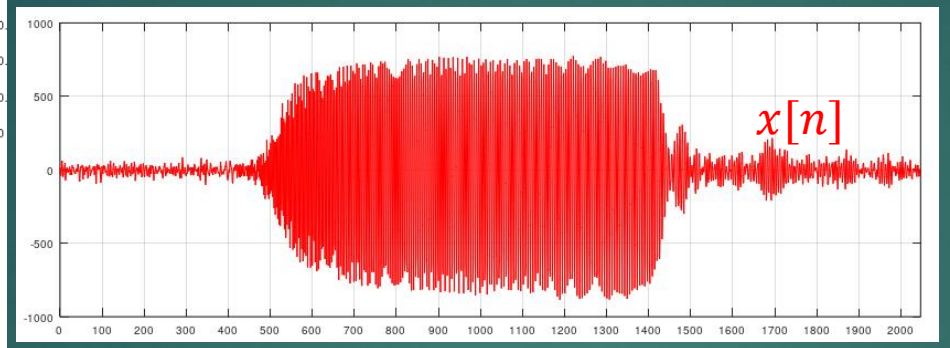


Qual a resposta do sistema $H(q) = \frac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ à entrada $x[n] = som\ do\ morcego$

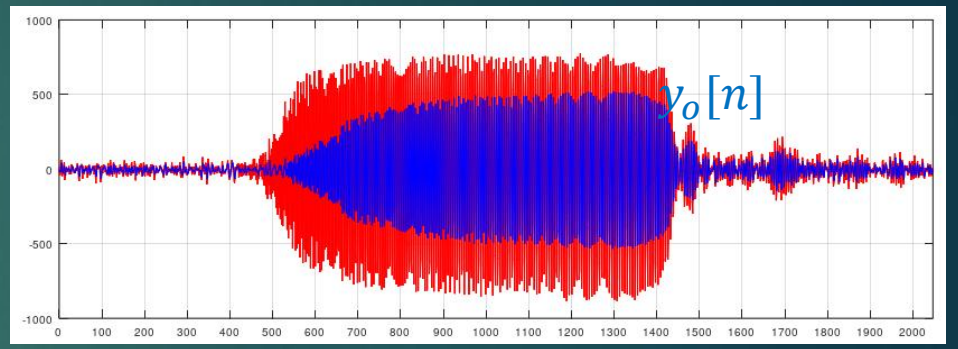
Numericamente por Equação de Diferenças

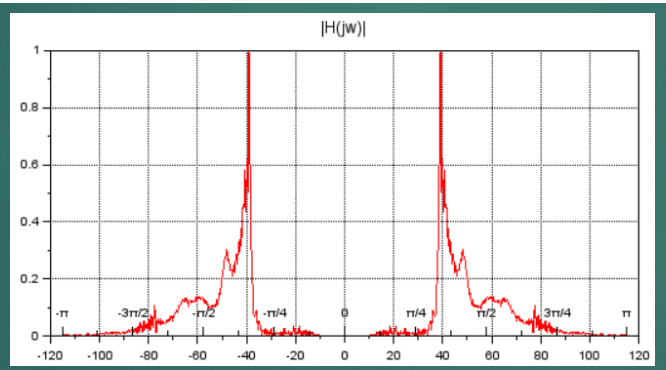


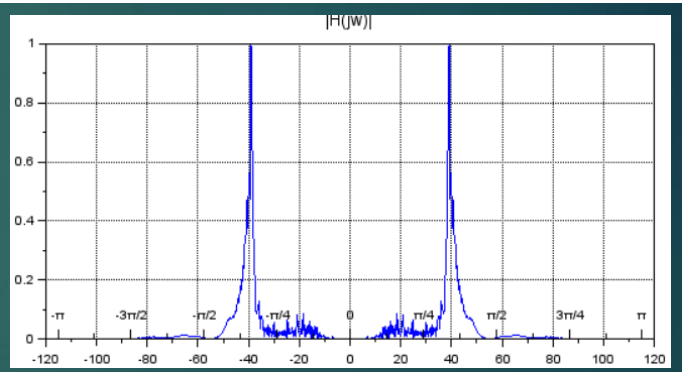

```
>> load('./Dados/bat.mat')
>> x_in=bat';
>> n=[0:length(x_in)-1];
>> plot(n,x_in,'r')
```

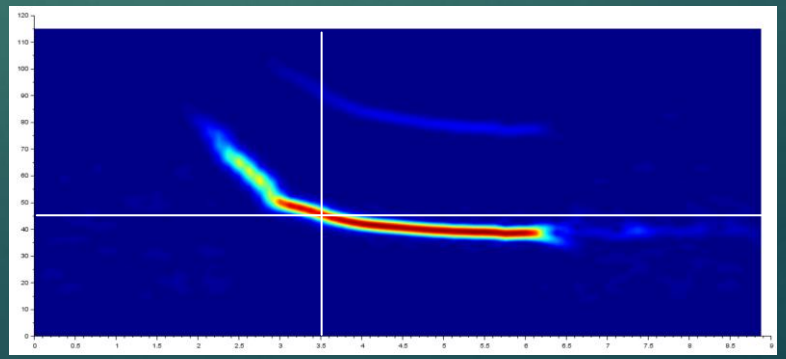



```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0]);
>> hold on; grid on
>> plot(n,x_in,'r')
>> plot(n,y_out,'b')
```



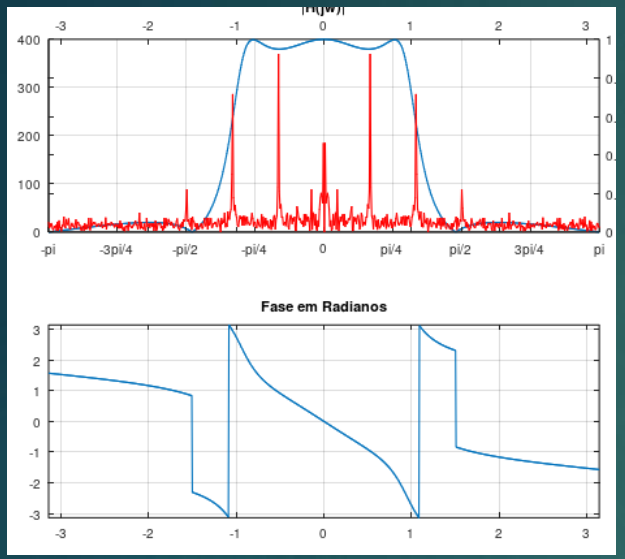
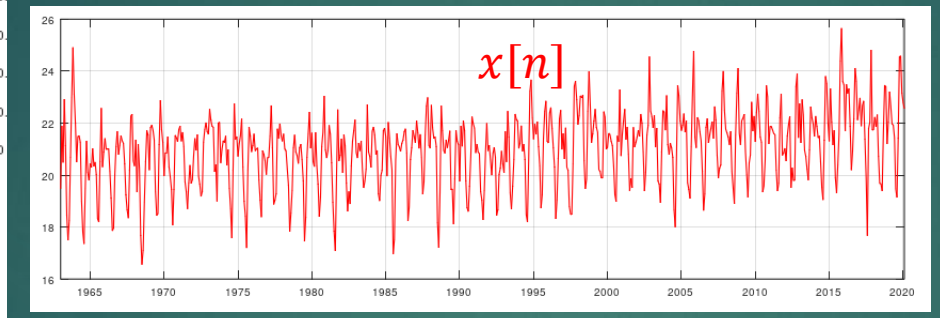
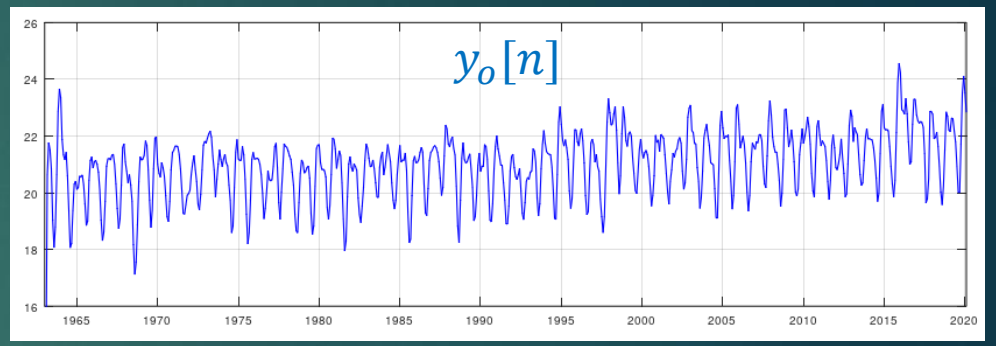
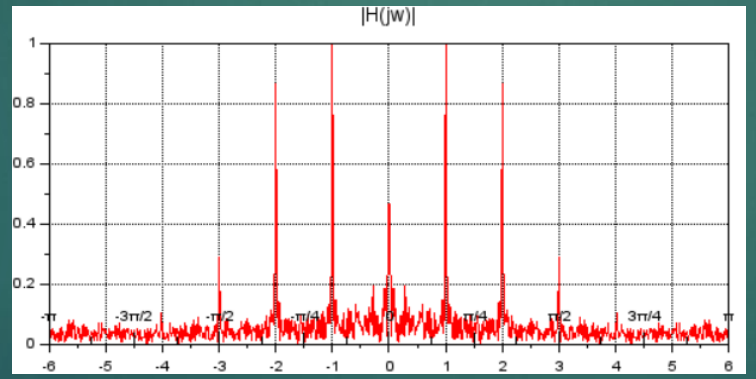
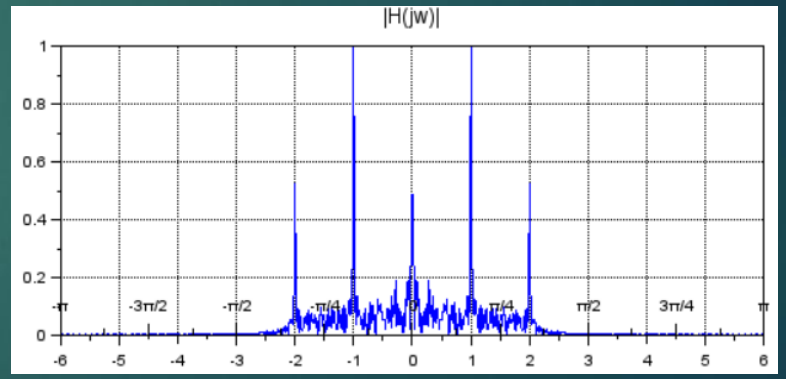






Qual a resposta do sistema $H(q) = \frac{\left(\frac{5}{49}\right)+\left(\frac{3}{34}\right)q+\left(\frac{3}{34}\right)q^2+\left(\frac{5}{49}\right)q^3}{1-\left(\frac{57}{40}\right)q+\left(\frac{37}{33}\right)q^2-\left(\frac{6}{19}\right)q^3}$ à entrada $x[n] = temperatura\ DF$

Numericamente por Equação de Diferenças

```
>> [n,x_in]=textread("./Dados/temperatura_DF_2.txt",
"%f %f",686);
>>
>> plot(n,x_in,'r')
```

```
>> Hz=1/3+1/3*z^-1+1/3*z^-2;
>> y_out=applyIIR(Hz,x_in,[0 0 0],[0 0 0]);
>> plot(n,y_out,'b')
```